Transcript for Demis Hassabis: Future of AI, Simulating Reality, Physics and Video Games | Lex Fridman Podcast #475

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Lex Fridman

Watch on

Episode highlight

Lex Fridman

(00:00:00) It's hard for us humans to make any kind of clean predictions about highly nonlinear dynamical systems. But again, to your point, we might be very surprised

what classical learning systems might be able to do about even fluid.

Demis Hassabis

(00:00:12) Yes, exactly. I mean, fluid dynamics, Navier-Stokes equations, these are traditionally thought of as very, very difficult intractable problems to do on classical systems. They take enormous amounts of compute, weather prediction systems. These kinds of things all involve fluid dynamics calculations.

(00:00:27) But again, if you look at something like Veo, our video generation model, it can model liquids quite well, surprisingly well. And materials, specular lighting, I love the ones where there's people who generate videos where there's clear liquids going through hydraulic presses and then it's being squeezed out. I used to write physics engines and graphics engines in my early days in gaming, and I know it's just so painstakingly hard to build programs that can do that. And yet somehow these systems are reverse engineering from just watching YouTube videos. So presumably what's happening is it's extracting some underlying structure around how these materials behave. So perhaps there is some kind of lower dimensional manifold that can be learned if we actually fully understood what's going on under the hood. That's maybe true of most of reality.

Introduction

Lex Fridman

(00:01:22) The following is a conversation with Demis Hassabis, his second time on the podcast. He is the leader of Google DeepMind and is now a Nobel Prize winner. Demis is one of the most brilliant and fascinating minds in the world today working on understanding and building intelligence and exploring the big mysteries of our universe. This was truly an honor and a pleasure for me.

(00:01:51) This is the Lex Fridman Podcast. To support it, please check out our sponsors in the description and consider subscribing to this channel. And now, dear friends, here's Demis Hassabis.

Learnable patterns in nature

Lex Fridman

(00:02:06) In your Nobel Prize lecture, you propose what I think is a super interesting conjecture that "any pattern that can be generated or found in nature can be efficiently discovered and modeled by a classical learning algorithm." What kind of patterns or systems might be included in that? Biology, chemistry, physics, maybe cosmology, neuroscience? What are we talking about?

Demis Hassabis

(00:02:32) Sure. Well, look, I felt that it's sort of a tradition, I think, of Nobel Prize lectures that you're supposed to be a little bit provocative and I wanted to follow that tradition. What I was talking about there is if you take a step back and you look at all the work that we've done, especially with the Alpha X projects, so I'm thinking AlphaGo, of course, AlphaFold, what they really are is we are building models of very combinatorially, high dimensional spaces that if you try to brute force a solution, find the best move and go, or find the exact shape of a protein, and if you enumerated all the possibilities, there wouldn't be enough time in the time of the universe.

(00:03:08) So you have to do something much smarter. And what we did in both cases was build models of those environments and that guided the search in a smart way and that makes it tractable. So if you think about protein folding, which is obviously a natural system, why should that be possible? How does physics do that? Proteins fold in milliseconds in our bodies, so somehow physics solves this problem that we've now also solved computationally. And I think the reason that's possible is that in nature, natural systems have structure because they were subject to evolutionary processes that shape them. And if that's true, then you can maybe learn what that structure is.

Lex Fridman

(00:03:49) This perspective I think is a really interesting one. You've hinted it at it, which is almost like crudely stated, anything that can be evolved can be efficiently modeled. Think there's some truth to that?

Demis Hassabis

(00:04:03) Yeah. I sometimes call it survival of the stablest or something like that because of course there's evolution for life, living things, but there's also, if you think about geological times, so the shape of mountains, that's been shaped by weathering processes over thousands of years, but then you can even take it cosmological, the orbits of planets, the shapes of asteroids. These have all been survived kind of processes

that have acted on them many, many times.

(00:04:31) If that's true, then there should be some sort of pattern that you can kind of reverse learn and a kind of manifold really that helps you search to the right solution, to the right shape and actually allow you to predict things about it in an efficient way because it's not a random pattern. So it may not be possible for man-made things or abstract things like factorizing large numbers because unless there's patterns in the number space, which there might be, but if there's not and it's uniform, then there's no pattern to learn, there's no model to learn that will help you search. So you have to do brute force. So in that case you maybe need a quantum computer, something like this. But in most things in nature that we're interested in are not like that. They have structure that evolved for a reason and survived over time. And if that's true, I think that's potentially learnable by a neural network.

Lex Fridman
(00:05:21) It's like nature is doing a search process and it's so fascinating that in that search process, it's creating systems that could be efficiently modeled.

Demis Hassabis
(00:05:31) That's right. Yeah.

Lex Fridman
(00:05:32) So interesting.

Demis Hassabis
(00:05:33) So they can be efficiently rediscovered or recovered because nature's not random. Everything that we see around us, including the elements that are more stable, all of those things, they're subject to some kind of selection process pressure.

Computation and P vs NP

Lex Fridman
(00:05:47) Do you think because you're also a fan of theoretical computer science and complexity, do you think we can come up with a complexity class, like a complexity zoo type of class where maybe it's the set of learnable systems, the set of learnable natural systems, LNS. This is a Demis Hassabis new class of systems that could be actually learnable by classical systems in this kind of way, natural systems that can be modeled

efficiently.

Demis Hassabis
(00:06:17) Yeah, I mean I've always been fascinated by the P equals NP question and what is model-able by classical systems, i.e. non-quantum systems, Turing machines in effect. And that's exactly what I'm working on actually in my few moments of spare time with a few colleagues about should there be maybe a new class or problem that is solvable by this type of neural network process and kind of mapped onto these natural systems, so the things that exist in physics and have structure. So I think that could be a very interesting new way of thinking about it. And it sort of fits with the way I think about physics in general, which is that I think information is primary, information is the most sort of fundamental unit of the universe, more fundamental than energy and matter. I think they can all be converted into each other, but I think of the universe as a kind of informational system.

Lex Fridman
(00:07:07) So when you think of the universe as an informational system, then the P equals NP question is a physics question.

Demis Hassabis
(00:07:14) That's right.

Lex Fridman
(00:07:15) And is a question that can help us actually solve the entirety of this whole thing going on.

Demis Hassabis
(00:07:20) Yeah, I think it's one of the most fundamental questions actually if you think of physics as informational and the answer to that, I think it's going to be very enlightening.

Lex Fridman
(00:07:29) More specific to the P and MP question, again, some of the stuff we're saying is kind of crazy right now just like the Christian Anfinsen Nobel Prize speech, controversial thing that he said sounded crazy and then you went and got a Nobel Prize for this with John Jumper, solved the problem. So let me just stick to the P equals NP. Do you think there's something in this thing we're talking about that could be shown if you can

do something like a polynomial time or constant time compute ahead of time and construct this gigantic model, then you can solve some of these extremely difficult problems in a theoretical computer science kind of way?

Demis Hassabis

(00:08:12) Yeah, I think that there are actually a huge class of problems that could be couched in this way, the way we did AlphaGo and the way we did AlphaFold, where you model what the dynamics of the system is, the properties of that system, the environment that you are trying to understand, and then that makes the search for the solution or the prediction of the next step efficient. Basically polynomial times, so tractable by a classical system, which a neural network is. It runs on normal computers, right? Classical computers, Turing machines in effect. And I think it's one of the most interesting questions there is, is how far can that paradigm go?

(00:08:53) I think we've proven, and the AI community in general that classical systems, Turing machines can go a lot further than we previously thought. They can do things like model the structures of proteins and play go to better than world champion level. And a lot of people would've thought maybe 10, 20 years ago that was decades away, or maybe you would need some sort of quantum machines to quantum systems to be able to do things like protein folding. And so I think we haven't really even sort of scratched the surface yet of what classical systems so-called could do.

(00:09:28) And of course, AGI being built on a neural network system on top of a neural network system on top of a classical computer would be the ultimate expression of that. And I think the limit, what the bounds of that kind of system, what it can do, it's a very interesting question and directly speaks to the P equals NP question.

Lex Fridman

(00:09:47) What do you think, again, hypothetical, might be outside of this? Maybe emergent phenomena? If you look at cellular automata, you have extremely simple systems and then some complexity emerges. Maybe that would be outside or even would you guess even that might be amenable to efficient modeling by a classical machine?

Demis Hassabis

(00:10:09) Yeah, I think those systems would be right on the boundary. So I think most emergent systems, cellular automata, things like that could be model-able by a classical system. You just sort of do a forward simulation of it and it'd probably be efficient

enough. Of course there's the question of things like chaotic systems where the initial conditions really matter and then you get to some uncorrelated end state. Now those could be difficult to model. So I think these are kind of the open questions, but I think when you step back and look at what we've done with the systems and the problems that we've solved, and then you look at things like Veo 3 on video generation sort of rendering physics and lighting and things like that, really core fundamental things in physics, it's pretty interesting. I think it's telling us something quite fundamental about how the universe is structured in my opinion. So in a way that's what I want to build AGI for is to help us as scientists answer these questions like P equals NP.

Lex Fridman

(00:11:09) Yeah, I think we might be continuously surprised about what is model-able by classical computers. I mean AlphaFold 3 on the interaction side is surprising that you can make any kind of progress on that direction. AlphaGenome is surprising that you can map the genetic code to the function. Kind of playing with the emergent kind of phenomena, you think there's so many combinatorial options and then here you go, you can find the kernel that is efficiently model-able.

Demis Hassabis

(00:11:36) Yes, because there's some structure, there's some landscape in the energy landscape or whatever it is that you can follow, some gradient you can follow. And of course what neural networks are very good at is following gradients. And so if there's one to follow and you can specify the objective function correctly, you don't have to deal with all that complexity, which I think is how we maybe have naively thought about it for decades, those problems. If you just enumerate all the possibilities, it looks totally intractable and there's many, many problems like that.

(00:12:06) And then you think, "Well, it's like 10 to 300 possible protein structures, 10 to the 170 possible go positions. All of these are way more than atoms in the universe, so how could one possibly find the right solution or predict the next step?" But it turns out that it is possible. And of course reality in nature does do it. Proteins do fold. So that gives you confidence that there must be, if we understood how physics was doing that in a sense and we could mimic that process, i.e. model that process, it should be possible on our classical systems is basically what the conjecture is about.

Lex Fridman

(00:12:44) And of course there's nonlinear dynamical systems, highly nonlinear dynamical systems, everything involving fluid. I recently had a conversation with Terence Tao who mathematically contends with a very difficult aspect of systems that have some singularities in them that break the mathematics, and it's just hard for us humans to make any kind of clean predictions about highly nonlinear dynamical systems. But again, to your point, we might be very surprised what classical learning systems might be able to do about even fluid.

Demis Hassabis

(00:13:16) Yes, exactly. I mean fluid dynamics, Navier-Stokes equations, these are traditionally thought of as very, very difficult, intractable kind of problems to do on classical systems. They take enormous amounts of compute, weather prediction systems. These kind of things all involve fluid dynamics calculations. But again, if you look at something like Veo, our video generation model, it can model liquids quite well, surprisingly well. And materials, specular lighting, I love the ones where there's people who generate videos where there's clear liquids going through hydraulic presses and then it's being squeezed out. I used to write physics engines and graphics engines in my early days in gaming, and I know it's just so painstakingly hard to build programs that can do that. And yet somehow these systems are reverse engineering from just watching YouTube videos. So presumably what's happening is it's extracting some underlying structure around how these materials behave. So perhaps there is some kind of lower dimensional manifold that can be learned if we actually fully understood what's going on under the hood. That's maybe true of most of reality.

Veo 3 and understanding reality

Lex Fridman

(00:14:26) Yeah, I've been continuously precisely by this aspect of Veo 3. I think a lot of people highlight different aspects including the comedic and the mean and all that kind of stuff. And then the ultra realistic ability to capture humans in a really nice way that's compelling and feels close to reality, and then combine that with native audio. All of those are marvelous things about Veo 3, but exactly the thing you're mentioning, which is the physics.

Demis Hassabis

(00:14:52) Yeah.

Lex Fridman

(00:14:53) It's not perfect, but it's pretty damn good. And then the really interesting scientific question is what is it understanding about our world in order to be able to do that? Because if the cynical, take with diffusion models, there's no way to understands anything. But I don't think you can generate that kind of video without understanding. And then our own philosophical notion of what it means to understand then is brought to the surface. To what degree do you think Veo 3 understands our world?

Demis Hassabis

(00:15:25) I think to the extent that it can predict the next frames in a coherent way, that is a form of understanding, not in the anthropomorphic version of, it's not some kind of deep philosophical understanding of what's going on, I don't think these systems have that, but they certainly have modeled enough of the dynamics, put it that way, that they can pretty accurately generate whatever it is, eight seconds of consistent video that by eye, at least at a glance, is quite hard to distinguish what the issues are.

(00:15:57) And imagine that in two or three more years' time, that's the thing I'm thinking about and how incredible that will look given where we've come from the early versions of that one or two years ago. And so the rate of progress is incredible. And I think I'm like you is like a lot of people love all of the stand-up comedians and that actually captures a lot of human dynamics very well and body language, but actually the thing I'm most impressed with and fascinated by is the physics behavior, the lighting and materials and liquids. And it's pretty amazing that it can do that. And I think that shows that it has some notion of at least intuitive physics, how things are supposed to work intuitively, maybe the way that a human child would understand physics, right, as opposed to a PhD student really being able to unpack all the equations. It's more of an intuitive physics understanding.

Lex Fridman

(00:16:53) Well, that intuitive physics understanding, that's the base layer, that's the thing people sometimes call a common-sense. It really understands something. I think that really surprised a lot of people. It blows my mind that I just didn't think it would be possible to generate that level of realism without understanding. There's this notion that you can only understand the physical world by having an embodied AI system, a

robot that interacts with that world. That's the only way to construct an understanding of that world. But Veo 3 is directly challenging that it feels like.

Demis Hassabis

(00:17:27) Yes, and it's very interesting, even if you were to ask me five, 10 years ago, I would've said, even though I was immersed in all of this, I would've said, "Well, yeah, you probably need to understand intuitive physics. If I push this off the table, this glass, it will maybe shatter and the liquid will spill out. So we know all of these things." But I thought that, and there's a lot of theories in neuroscience, it's called action in perception where you need to act in the world to really, truly perceive it in a deep way. And there was a lot of theories about you'd need embodied intelligence or robotics or something, or maybe at least simulated action so that you would understand things like intuitive physics.

(00:18:06) But it seems like you can understand it through passive observation, which is pretty surprising to me. And again, I think hints at something underlying about the nature of reality in my opinion, beyond just the cool videos that it generates. And of course there's next stages is maybe even making those videos interactive so one can actually step into them and move around them, which would be really mind-blowing, especially given my games background. So you can imagine. And then I think we're starting to get towards what I would call a world model, a model of how the world works, the mechanics of the world, the physics of the world, and the things in that world. And of course that's what you would need for a true AGI system.

Video games

Lex Fridman

(00:18:50) I have to talk to you about video games. So you are being a bit trolley. I think you're having more and more fun on Twitter, on X, which is great to see. So a guy named Jimmy Apples tweeted, "Let me play a video game of my Veo 3 videos already. Google cooked so good. Playable world models wen?" And then you co-tweeted that with, "Now, wouldn't that be something?" So how hard is it to build game worlds with AI? Maybe can you look out into the future feature of video games five, 10 years out? What do you think that looks like?

Demis Hassabis

(00:19:27) Well, games were my first love really. And doing AI for games was the first thing I did professionally in my teenage years and with the first major AI systems that I built and I always want to scratch that itch one day and come back to that. And I will do, I think, and I think I sort of dream about what would I have done back in the nineties if I'd had access to the kind of AI systems we have today? And I think you could build absolutely mind-blowing games.

(00:19:55) And I think the next stage is I always used to love making, all the games I've made are open world games, so they're games where there's a simulation and then there's AI characters, and then the player interacts with that simulation and the simulation adapts to the way the player plays. And I always thought they were the coolest games because, so games like Theme Park that I worked on where everybody's game experience would be unique to them because you're kind of co-creating the game. We set up the parameters, we set up initial conditions, and then you as the player immersed in it, and then you are co-creating it with the simulation. But of course it's very hard to program open world games. You've got to be able to create content whichever direction the player goes in, and you want it to be compelling no matter what the player chooses. And so it was always quite difficult to build things like cellular automata actually, type of those kind of classical systems, which created some emergent behavior, but they're always a little bit fragile, a little bit limited. Now we are maybe on the cusp in the next few years, five, 10 years of having AI systems that can truly create around your imagination, can dynamically change the story and storytell the narrative around and make it dramatic no matter what you end up choosing. So it's like the ultimate choose your own adventure sort of game. And I think maybe we are within reach, if you think of a kind of interactive version of Veo and then wind that forward five to 10 years and imagine how good it's going to be.

Lex Fridman

(00:21:24) Yeah. So you said a lot of super interesting stuff there. So one, the open world, built into that is a deep personalization the way you've described it. So it's not just that it's open world, that you can open any door and there'll be something there, it's that it's the choice of which door you open in an unconstrained way defines the worlds you see. So some games try to do that, they give you choice, but it's really just an illusion of choice because you only, like Stanley Parable, a game I actually played, it's really, there's a couple of doors and it really just takes you down a narrative. Stanley

Parable is a great video game. I recommend people play it, that in a meta way, mocks the illusion of choice, and there's philosophical notions of free will and so on.

(00:22:12) But I do, one of my favorite games of Elder Scrolls is Daggerfall I believe, that they really played with a random generation of the dungeons of if you can step in and they give you this feeling of an open world. And there you mentioned interactivity. You don't need to interact. That's the first step because you don't need to interact that much. You just, when you open the door, whatever you see is randomly generated for you. And that's already an incredible experience because you might be the only person to ever see that.

Demis Hassabis

(00:22:46) Yeah, exactly. But what you'd like is a little bit better than just sort of a random generation. So you'd like, and also better than a simple AB hard coded choice, right? That's not really open world, as you say. It's just giving you the illusion of choice. What you want to be able to do is potentially anything in that game environment. And I think the only way you can do that is to have generated systems, systems that will generate that on the fly. Of course, you can't create infinite amounts of game assets. It's expensive enough already how AAA games are made today. And that was obvious to us back in the nineties when I was working on all these games.

(00:23:25) I think maybe Black & White was the game that I worked on early stages of that, that had still probably the best AI, learning AI, in it. It was an early reinforcement learning system that you were looking after this mythical creature and growing it and nurturing it. And depending how you treated it, it would treat the villagers in that world the same way. So if you were mean to it, it would be mean. If you were good, it would be protective. And so it was really a reflection of the way you played it. So actually all of the, I've been working on simulations and AI through the medium of games at the beginning of my career, and really the whole of what I do today, it's still a follow on from those early more hard coded ways of doing the AI to now fully general learning systems that are trying to achieve the same thing.

Lex Fridman

(00:24:12) Yeah, it is been interesting, hilarious, and fun to watch you and Elon obviously itching to create games because you're both gamers. And one of the sad aspects of your incredible success in so many domains of science, like serious adult stuff, that you

might not have time to really create a game, you might end up creating the tooling that others will create the game. You have to watch others create the thing you've always dreamed of. Do you think it's possible you can somehow in your extremely busy schedule actually find time to create something like Black & White, an actual video game where you could make the childhood dream become reality?

Demis Hassabis

(00:24:58) Well, there's two things where I think about that is maybe with vibe coding as it gets better and there's a possibility that I could, one could do that actually in your spare time. So I'm quite excited about that as that would be my project if I got the time to do some vibe coding. I'm actually itching to do that. And then the other thing is maybe it's a sabbatical after AGI has been safely stewarded into the world and delivered into the world. That, and then working on my physics theory as we talked about at the beginning, those would be my two post-AGI projects, let's call it that way.

Lex Fridman

(00:25:28) I would love to see which you choose, solving the problem that some of the smartest people in human history contended with, P equals NP, or creating a cool video game.

Demis Hassabis

(00:25:44) But in my world, they'd be related because it would be an open world simulated game as realistic as possible. So what is the universe? That's speaking to the same question and P equals NP. I think all these things are related, at least in my mind.

Lex Fridman

(00:25:59) I mean in a really serious way, video games sometimes are looked down upon as just this fun side activity. But especially as AI does more and more of the difficult, boring tasks, something we in modern world called work, video games is the thing in which we may find meaning, in which we may find what to do with our time. You could create incredibly rich, meaningful experiences. That's what human life is. And then in video games, you can create more sophisticated, more diverse ways of living. Right? That's the point?

Demis Hassabis

(00:26:41) I think so. I mean, those of us who love games and I still do is it's almost can let your imagination run wild, right? I used to love games and working on games so

much because it's the fusion, especially in the nineties and early two thousands, the sort of golden era and maybe the eighties of the games industry. And it was all being discovered. New genres were being discovered. We weren't just making games, we felt we were creating a new entertainment medium that never existed before. Especially with these open world games and simulation games where you, as the player, were co-creating the story. There's no other media, entertainment media, where you do that, where you as the audience actually co-create the story.

(00:27:23) And of course now with multiplayer games as well, it can be a very social activity and can explore all kinds of interesting worlds in that. But on the other hand, it's very important to also enjoy and experience the physical world. But the question is then I think we're going to have to confront the question again of what is the fundamental nature of reality? What is going to be the difference between these increasingly realistic simulations and multiplayer ones and emergent and what we do in the real world?

Lex Fridman
(00:27:55) Yeah, there's clearly a huge amount of value to experiencing the real world, nature. There's also a huge amount of value in experiencing other humans directly in person the way we're sitting here today, but we need to really scientifically rigorously answer the question why and which aspect of that can be mapped into the virtual world.

Demis Hassabis
(00:28:17) Exactly.

Lex Fridman
(00:28:18) It's not enough to say, "Yeah, you should go touch grass and hang out in nature." It's like why exactly is that valuable?

Demis Hassabis
(00:28:25) Yes. And I guess that's maybe the thing that's been haunting me or obsessing me from the beginning of my career. If you think about all the different things I've done, they're all related in that way. The simulation, nature of reality, and what is the bounds of what can be modeled.

Lex Fridman

(00:28:40) Sorry for the ridiculous question, but so far, what is the greatest video game of all time? What's up there?

Demis Hassabis
(00:28:45) Well, my favorite one of all time is Civilization, I have to say. That was the Civilization I and Civilization II, my favorite games of all time.

Lex Fridman
(00:28:54) I can only assume you've avoided the most recent one because it would probably, that would be your sabbatical. You would disappear.

Demis Hassabis
(00:29:03) Yes, exactly. They take a lot of time, these Civilization games, so I've got to be careful with them.

Lex Fridman
(00:29:08) Fun question. You and Elon seem to be somehow solid gamers. Is there a connection between being great at gaming and being great leaders of AI companies?

Demis Hassabis
(00:29:20) I don't know. It's an interesting one. I mean, we both love games and it's interesting, he wrote games as well to start off with. Probably, especially in the era I grew up in where home computers just became a thing in the late eighties and nineties, especially in the UK, I had a spectrum and then a Commodore Amiga 500, which was my favorite computer ever. And that's why I learned all my programming. And of course, it's a very fun thing to program, is to program games. So I think it's a great way to learn programming, probably still is. And then of course, I immediately took it in directions of AI and simulations, so I was able to express my interest in games and my wider scientific interests all together.

Demis Hassabis
(00:30:00) And my sort of wider scientific interests all together. And then the final thing I think that's great about games is it fuses artistic design, art, with the most cutting edge programming. So again, in the nineties, all of the most interesting technical advances were happening in gaming, whether that was AI, graphics, physics engines, hardware, even GPUs of course were designed for gaming originally. So everything that was pushing computing forward in the nineties was due to gaming. So interestingly,

that was where the forefront of research was going on and it was this incredible fusion with art. Graphics, but also music, and just the whole new media of storytelling. And I love that. For me, it's this sort of multidisciplinary kind of effort is again something I've enjoyed my whole life.

AlphaEvolve

Lex Fridman

(00:30:52) I have to ask you, I almost forgot about one of the many, and I would say one of the most incredible things recently that somehow didn't yet get enough attention is AlphaEvolve. We talked about Evolution a little bit, but it's the Google DeepMind system that evolves algorithms. Are these kinds of Evolution-like techniques promising as a component of future super intelligence systems? So for people who don't know, it's kind of, I don't know if it's fair to say it's LLM guided Evolution search because Evolution algorithms are doing the search and LLMs are telling you where.

Demis Hassabis

(00:31:28) Yes. Yes, exactly. So LLMs are kind of proposing some possible solutions and then you use evolutionary computing on top to find some novel part of the search space. So actually I think it's an example of very promising directions where you combine LLMs or foundation models with other computational techniques. Evolutionary methods is one, but you could also imagine Monte Carlo tree search. Basically many types of search algorithms or reasoning algorithms sort of on top of or using the foundation models as a basis. So I actually think there's quite a lot of interesting things to be discovered probably with these sort of hybrid systems, let's call them.

Lex Fridman

(00:32:10) But not to romanticize Evolution.

Demis Hassabis

(00:32:12) Yeah.

Lex Fridman

(00:32:12) I'm only human, but you think there's some value in whatever that mechanism is? Because we already talked about natural systems. Do you think where there's a lot of low-hanging fruit of us understanding, being able to model, being able to simu-

late Evolution and then using that, whatever we understand about that nature-inspired mechanism, to then do search better and better and better?

Demis Hassabis

(00:32:36) Yes. So if you think about, again, breaking down the solar systems we've built to their really fundamental core, you've got the model of the underlying dynamics of the system. And then if you want to discover something new, something novel that hasn't been seen before, then you need some kind of search process on top to take you to a novel of the search space. And you can do that in a number of ways. Evolutionary computing is one. With AlphaGo, we just use Monte Carlo Tree Search and that's what found move 37, the new never seen before strategy in Go. And so that's how you can go beyond potentially what is already known. So the model can model everything that you currently know about, all the data that you currently have. But then how do you go beyond that? So that starts to speak about the ideas of creativity.

(00:33:28) How can these systems create something new? In fact discover something new? Obviously this is super relevant for scientific discovery or pushing met science and medicine forward, which we want to do with these systems. And you can actually bolt on some fairly simple search systems on top of these models and get you into a new region of space. Of course, you also have to make sure that you're not searching that space totally randomly. It would be too big. So you have to have some objective function that you're trying to optimize and hill climb towards and that guides that search.

Lex Fridman

(00:34:00) But there's some mechanism of Evolution that are interesting maybe in the space of programs. But then the space of programs an extremely important space, because you can probably generalize to everything. But for example, mutation. So it's not just Monte Carlo Tree Search where it's like a search. You could every once in a while-

Demis Hassabis

(00:34:21) Combine things.

Lex Fridman

(00:34:22) Combine things?

Demis Hassabis

(00:34:23) Yeah.

Lex Fridman
(00:34:23) Things, like the components of a thing.

Demis Hassabis
(00:34:26) Yes.

Lex Fridman
(00:34:26) So then what Evolution is really good at is not just the natural selection, it's combining things and building increasingly complex hierarchical systems. So that component is super interesting, especially with AlphaEvolve and the space of programs.

Demis Hassabis
(00:34:43) Yeah, exactly. So you can get a bit of an extra property out of evolutionary systems, which is some new emerging capability may come about, right? Of course like happened with life, interestingly with naive, traditional evolutionary computing methods without LLMs and the modern AI, the problem with them, they were very well studied in the nineties and early two thousands and some promising results, but the problem was they could never work out how to evolve new properties, new emerging properties. You always had a sort of subset of the properties that you put into the system, but maybe if we combine them with these foundation models, perhaps we can overcome that limitation.

(00:35:21) Obviously naturally evolution clearly did. It did evolve new capabilities. So bacteria to where we are now. So clearly that it must be possible with evolutionary systems to generate new patterns, going back to the first thing we talked about and new capabilities and emerging properties, and maybe we're on the cusp of discovering how to do that.

Lex Fridman
(00:35:44) Yeah, listen, AlphaEvolve is one of the coolest things I've ever seen. I've on my desk at home, most of my time is spent on that computer is just programming. And next to the three screens is a skull of a Tiktaalik, which is one of the early organisms that crawled out of the water onto land. And I just kind of watch that little guy. It's like whatever the competition mechanism of Evolution is, it's quite incredible.

Demis Hassabis

(00:36:16) Yes.

Lex Fridman

(00:36:17) It's truly, truly incredible. Now whether that's exactly the thing we need to do to do our search, but never dismiss the power of nature, what it did here.

Demis Hassabis

(00:36:27) And it's amazing, which is a relatively simple algorithm, right? Effectively, and it can generate all of this immense complexity emerges obviously running over 4 billion years of time. But you can think about that as again, a search process that ran over the physics substrate of the universe for a long amount of computational time, but then it generated all this incredible rich diversity.

AI research

Lex Fridman

(00:36:54) So many questions I want to ask you. So one, you do have a dream, one of the natural systems you want to try to model is a cell. That's a beautiful dream. I could ask you about that. I also just for that purpose on the AI scientist front just broadly, so there's a essay from Daniel Cocotaglio, Scott Alexander and others that online steps along the way to get to ASI and it has a lot of interesting ideas in it, one of which is including a superhuman coder and a superhuman AI researcher. And in that there's a term of research taste that's really interesting. So in everything you've seen, do you think it's possible for AI systems to have research taste to help you in the way that AI co-scientist does, to help steer human brilliant scientists and then potentially by itself to figure out what are the directions where you want to generate truly novel ideas? That seems to be a really important component of how to do great science?

Demis Hassabis

(00:38:03) Yeah, I think that's going to be one of the hardest things to mimic or model is this idea of taste or judgment. I think that's what separates the great scientists from the good scientists. All professional scientists are good technically, otherwise they wouldn't have made it that far in academia and things like that. But then do you have the taste to sniff out what the right direction is, what the right experiment is, what the right question is? So picking the right question is the hardest part of science and making the right hypothesis. And that's what today's systems definitely they can't do. So I often say

it's harder to come up with a conjecture, a really good conjecture than it is to solve it. So we may have systems soon that can solve pretty hard conjectures. A maths Olympiad problems, where Alpha Proof last year our system got silver medal in that really hard problems. Maybe eventually we'll better solve a Millennium Prize kind of problem. But could a system have come up with a conjecture worthy of study that someone like Terence Tao would've gone? "You know what, that's a really deep question about the nature of maths or the nature of numbers or the nature of physics." And that is far harder type of creativity. And we don't really know. Today's systems clearly can't do that. And we're not quite sure what that mechanism would be. This kind of leap of imagination like Einstein had when he came up with special relativity and then general relativity with the knowledge he had at the time.

Lex Fridman

(00:39:33) For conjecture, you want to come up with a thing that's interesting, it's amenable to proof?

Demis Hassabis

(00:39:41) Yes.

Lex Fridman

(00:39:42) So it's easy to come up with a thing that's extremely difficult. It's easy to come up with a thing that's extremely easy, but at that very edge-

Demis Hassabis

(00:39:48) That sweet spot of basically advancing the science and splitting the hypothesis space into two, ideally. Right? Whether if it's true or not true, you've learned something really useful and that's hard. And making something that's also falsifiable and within the technologies that you currently have available. So it's a very creative process, actually. A highly creative process that I think just a kind of naive search on top of a model won't be enough for that.

Lex Fridman

(00:40:21) The idea of splitting the hypothesis space in two is super interesting. So I've heard you say that there's basically no failure or failure is extremely valuable if you construct the questions right, if you construct the experiments right, if you design them right, that failure or success are both useful, so perhaps because it splits the hypothesis basically in two, it's like a binary search?

Demis Hassabis

(00:40:44) Yes, that's right. So when you do real Blue Sky research, there's no such thing as failure really. As long as you are picking experiments and hypotheses that meaningfully split the hypothesis space and you learn something. You can learn something kind of equally valuable from an experiment that doesn't work. That should tell you if you've designed the experiment well and your hypotheses are interesting, it should tell you a lot about where to go next. And then you're effectively doing a search process and using that information in very helpful ways.

Simulating a biological organism

Lex Fridman

(00:41:17) So to go to your dream of modeling a cell, what are the big challenges that lay ahead for us to make that happen? We should maybe highlight that in AlphaFold, I mean there's just so many leaps. So AlphaFold solved, if it's fair to say, protein folding. And there's so many incredible things we could talk about there, including the open sourcing, everything you've released AlphaFold 3 is doing protein, RNA, DNA interactions, which is super complicated and fascinating. It's amenable to modeling. AlphaGenome predicts how small genetic changes if we think about single mutations, how they link to actual function. So it seems like it's creeping along to sophisticated to much more complicated things like a cell. But a cell has a lot of really complicated components.

Demis Hassabis

(00:42:11) So what I've tried to do throughout my career is I have these really grand dreams and then I try to, as you've noticed, but I try to break them down. It's easy to have a kind of crazily ambitious dream, but the trick is how do you break it down into manageable, achievable, interim steps that are meaningful and useful in their own right? And so Virtual Cell, which is what I call the project of modeling a cell, I've had this idea of wanting to do that for maybe more like 25 years.

(00:42:41) And I used to talk with Paul Nurse, who is a bit of a mentor of mine in biology. He runs the founded the Crick Institute and won the Nobel Prize in 2001. We've been talking about it since the nineties, and I used to come back to it every five years. It's like, what would you need to model the full internals of a cell so that you could do

experiments on the virtual cell and what those experiment in silico and those predictions would be useful for you to save you a lot of time in the wet lab. That would be the dream.

(00:43:13) Maybe you could a hundred x speed up experiments by doing most of it in silico the search in silico, and then you do the validation step in the wet lab. That's the dream. But maybe now, finally, so I was trying to build these components, AlphaFold being one, that would allow you eventually to model the full interaction, a full simulation of a cell, and I'd probably start with a yeast cell. And partly that's what Paul Nurse studied because the yeast cell is like a full organism, that's a single cell. So it's the kind of simplest single cell organism. And so it's not just a cell, it's a full organism.

(00:43:49) And yeast is very well understood. And so that would be a good candidate for a kind of full simulated model. Now AlphaFold is the solution to the kind of static picture of what does a 3D structure protein look like? A static picture of it. But we know that biology, all the interesting things happen with the dynamics, the interactions, and that's what AlphaFold 3 is, the first step towards is modeling those interactions. So first of all, pair wise proteins with proteins, proteins with RNA and DNA. But then the next step after that would be modeling maybe a whole pathway, maybe like the tour pathway that's involved in cancer or something like this. And then eventually you might be able to model a whole cell.

Lex Fridman

(00:44:31) Also, there's another complexity here that stuff in a cell happens at different time scales. Is that tricky? Protein folding is super fast. I don't know all the biological mechanisms, but some of them take a long time. And so the levels of interaction has a different temporal scale that you have to be able to model.

Demis Hassabis

(00:44:54) So that would be hard. So you'd probably need several simulated systems that can interact at these different temporal dynamics, or at least maybe it's like a hierarchical system so you can jump up or down the different temporal stages.

Lex Fridman

(00:45:07) So can you avoid... One of the challenges here is not avoid simulating, for example, the quantum mechanical aspects of any of this, right? You want to not over model. You can skip ahead to just model the really high level things that get you a really

good estimate of what's going to happen.

Demis Hassabis

(00:45:27) Yes. So you got to make a decision when you're modeling any natural system, what is the cutoff level of the granularity that you're going to model it to? And then it captures the dynamics that you're interested in. So probably for a cell I would hope that would be the protein level, and that one wouldn't have to go down to the atomic level. So of course that's where AlphaFold stock kicks in. So that would be kind of the basis, and then you'd build these higher level simulations that take those as building blocks and then you get the emergent behavior.

Origin of life

Lex Fridman

(00:46:00) I apologize for the pothead questions ahead of time, but do you think we'll be able to simulate a model, the origin of life? So being able to simulate the first from non-living organisms, the birth of a living organism?

Demis Hassabis

(00:46:19) I think that's one of course one of the deepest and most fascinating questions. I love that area of biology. There's people, there's a great book by Nick Lane, one of the top experts in this area called The Ten Great Inventions of Evolution. I think it's fantastic. And it also speaks to what the great filters might be, prior or are they ahead of us? I think they're most likely in the past, if you read that book of how unlikely to go have any life at all. And then single cell to multi-cell seems an unbelievably big jump that took a billion years, I think on earth to do, right? So it shows you how hard it was.

Lex Fridman

(00:46:55) Right? Bacteria were super happy for a very long time.

Demis Hassabis

(00:46:56) For a very long time before they captured mitochondria somehow, right? I don't see why not, why AI couldn't help with that. Some kind of simulation. Again, it's a bit of a search process through a combinatorial space. Here's all the chemical soup that you start with, the primordial soup, that maybe was on earth near these hot vents. Here's some initial conditions. Can you generate something that looks like a cell? So

perhaps that would be a next stage after the virtual cell project is well, how could something like that emerge from the chemical soup?

Lex Fridman
(00:47:31) Well, I would love it if there was a Move 37 for the origin of life. I think that's one of the great mysteries. I think ultimately what we'll figure out is their continuum. There's no such thing as a line between non-living and living. But if we can make that rigorous.

Demis Hassabis
(00:47:44) Yes.

Lex Fridman
(00:47:45) That the very thing from the Big Bang to today has been the same process. If you can break down that wall that we've constructed in our minds of the actual origin from non-living to living, and it's not a line that it's a continuum that connects physics and chemistry and biology. There's no line.

Demis Hassabis
(00:48:03) I mean, this is my whole reason why I worked on AI and AGI my whole life, because I think it can be the ultimate tool to help us answer these kinds of questions. And I don't really understand why the average person doesn't worry about this stuff more. How can we not have a good definition of life and not living a non-living and the nature of time and let alone consciousness and gravity and all these things and quantum mechanics weirdness? It's just to me, I've always had this sort of screaming at me in my face and it's getting louder. It's like, what is going on here? And I mean that in the deeper sense, the nature of reality, which has to be the ultimate question that would answer all of these things. It's sort of crazy if you think about it. We can stare at each other and all these living things all the time. We can inspect it microscopes and take it apart almost down to the atomic level. And yet we still can't answer that clearly in a simple way. That question of how do you define living? It's kind of amazing.

Lex Fridman
(00:49:05) Yeah, living, you can kind of talk your way out of thinking about. But consciousness, we have this very obviously subjective, conscious experience like we're at the center of our own world and feels like something. And then how are you not screaming at the mystery of it all? I mean, but really humans have been contending

with the mystery of the world around them for a long, long... There's a lot of mysteries like what's up with the sun and the rain? What's that about? And then last year we had a lot of rain, and this year we don't have rain. What did we do wrong? Humans have been asking that question for a long time.

Demis Hassabis
(00:49:46) Exactly. So I guess we've developed a lot of mechanisms to cope with these deep mysteries that we can't fully, we can see, but we can't fully understand and we have to just get on with daily life. And we keep ourselves busy in a way. In a way, did we keep ourselves distracted?

Lex Fridman
(00:50:01) I mean, weather is one of the most important questions of human history. We still, that's the go-to small talk direction of the weather.

Demis Hassabis
(00:50:09) Yes. Especially in England.

Lex Fridman
(00:50:11) And then which is famously is an extremely difficult system to model. And even that system, Google DeepMind has made progress on.

Demis Hassabis
(00:50:22) Yes, we've created the best weather prediction systems in the world and they're better than traditional fluid dynamics sort of systems that usually calculated on massive supercomputers takes days to calculate it. And we've managed to model a lot of the weather dynamics with neural network systems, with our WeatherNet system. And again, it's interesting that those kinds of dynamics can be modeled even though very complicated, almost bordering on chaotic systems in some cases.

(00:50:50) A lot of the interesting aspects of that can be modeled by these neural network systems, including very recently we had cyclone prediction of where paths of hurricanes might go. Of course, super useful, super important for the world and it's super important to do that very timely and very quickly and as well as accurately. And I think it's very promising direction again, of simulating so that you can run forward predictions and simulations of very complicated real world systems.

Lex Fridman

(00:51:18) I should mention that I've gotten a chance in Texas to meet a community of folks called the Storm Chasers. And what's really incredible about them, I need to talk to them more, is they're extremely tech-savvy because what they have to do is they have to use models to predict where the storm is. So it's this beautiful mix of crazy enough to go into the eye of the storm and in order to protect your life and predict where the extreme events are going to be, they have to have increasingly sophisticated models of weather.

Demis Hassabis
(00:51:50) Yeah.

Lex Fridman
(00:51:50) It is a beautiful balance of being in it as living organisms and the cutting edge of science. They actually might be using DeepMind systems.

Demis Hassabis
(00:52:01) Yeah. But hopefully they are. And I'd love to join them in one of those checks. They look amazing. Right. That's great to actually experience it one time.

Path to AGI

Lex Fridman
(00:52:07) Exactly. And then also to experience the correct prediction where something will come and how it's going to evolve. It's incredible. You've estimated that we'll have AGI by 2030, so there's interesting questions around that. How will we actually know that we got there and what may be the move quote, "Move 37" of AGI.

Demis Hassabis
(00:52:33) My estimate is sort of 50% chance by in the next five years, so by 2030 let's say. So I think there's a good chance that that could happen. Part of it is what is your definition of AGI? Of course people arguing about that now and mind's quite a high bar and always has been of can we match the cognitive functions that the brain has? So we know our brains are pretty much general Turing machines approximate, and of course we created incredible modern civilization with our minds. So that also speaks to how general the brain is.

(00:53:06) And for us to know we have a true AGI, we would have to make sure that it

has all those capabilities. It isn't kind of a jagged intelligence where some things, it's really good at, like today's systems, but other things it's really flawed at. And that's what we currently have with today's systems. They're not consistent. So you'd want that consistency of intelligence across the board.

(00:53:28) And then we have some missing, I think, capabilities like the true invention capabilities and creativity that we were talking about earlier. So you'd want to see those. How you test that? I think you just test it. One way to do it would be kind of brute force test of tens of thousands of cognitive tasks that we know that humans can do. And maybe also make the system available to a few hundred of the world's top experts, the Terence Taos of each subject area and give them a month or two and see if they can find an obvious flaw in the system. And if they can't, then I think you can be pretty confident we have a fully general system.

Lex Fridman

(00:54:11) Maybe to push back a little bit, it seems like humans are really incredible as the intelligence improves across all domains to take it for granted, like you mentioned, Terence Tao, these brilliant experts. They might quickly in a span of weeks, take for granted all the incredible things it can do and then focus in on, well, aha, right there. I consider myself, first of all, human. I identify as human. Some people listen to me talk and they're like, "That guy is not good at talking the stuttering." So even humans have obvious across domains limits, even just outside of calc, mathematics and physics and so on. I wonder if it will take something like a Move 37, so on the positive side versus a barrage of 10,000 cognitive tasks where it'll be one or two where it's like, holy shit, this is special.

Demis Hassabis

(00:55:14) So I think that. Exactly. So I think there's the sort of blanket testing to just make sure you've got the consistency. But I think there are the sort of lighthouse moments like the Move 37 that I would be looking for. So one would be inventing a new conjecture or a new hypothesis about physics like Einstein did.

(00:55:33) So maybe you could even run the back test of that very rigorously, have a cut-off of 1900 and then give the system everything that was written up to 1900 and then see if it could come up with special relativity and general relativity, right? Like Einstein did. That would be an interesting test. Another one would be can it invent a game

like Go? Go not just come up with Move 37, a new strategy, but can it invent a game that's as deep as aesthetically beautiful, as elegant as Go? And those are the sorts of things I would be looking out for. And probably a system being able to do several of those things for it to be very general, not just one domain. And so I think that would be the signs at least that I would be looking for, that we've got a system that's AGI level and then maybe to fill that out, you would also check their consistency, make sure there's no holes in that system either.

Lex Fridman
(00:56:27) Yeah, something like a new conjecture or scientific discovery. That would be a cool feeling.

Demis Hassabis
(00:56:32) Yeah, that would be amazing. So it's not just helping us do that, but actually coming up with something brand new.

Lex Fridman
(00:56:38) And you would be in the room for that.

Demis Hassabis
(00:56:40) Absolutely.

Lex Fridman
(00:56:40) It would be probably two or three months before announcing it. And you would just be sitting there trying not to Tweet.

Demis Hassabis
(00:56:49) Something like that. Exactly. It's like what is this amazing new physics idea? And then we would probably check it with world experts in that domain and validate it and go through its workings. And I guess it would be explaining its workings too. Yeah. It would be an amazing moment.

Lex Fridman
(00:57:07) Do you worry that we as humans, even expert humans, like you might miss it? Might miss-

Demis Hassabis
(00:57:12) Well, it may be pretty complicated. So it could be, the analogy I give there is I don't think it will be totally mysterious to the best human scientists, but it may be a bit

like, for example in chess, if I was to talk to Garry Kasparov for Magnus Carlsen and play a game with them and they make a brilliant move, I might not be able to come up with that move. But they could explain why afterwards that move made sense. And we would be to understand it to some degree, not to the level they do, but if they were good at explaining, which is actually part of intelligence too, is being able to explain in a simple way that what you're thinking about, I think that that will be very possible for the best human scientists.

Lex Fridman

(00:57:52) But I wonder, maybe you can educate me on the side of Go, I wonder if there's moves from Magnus or Garry where they at first will dismiss it as a bad move?

Demis Hassabis

(00:58:02) Yeah, sure, it could be. But then afterwards they'll figure out with their intuition why this works. And then empirically, the nice thing about games is, one of the great things about games is it's a sort of scientific test. Do you win the game or not win? And then that tells you, okay, that move in the end was good, that strategy was good. And then you can go back and analyze that and explain even to yourself a little bit more why. Explore around it, and that's how chess analysis and things like that works. So perhaps that's why my brain works like that because I've been doing that since I was four and it's sort of hardcore training in that way.

Lex Fridman

(00:58:39) But even now when I generate code, there is this kind of nuanced, fascinating contention that's happening where I might at first identify as a set of generated code is incorrect in some interesting nuanced ways. But then I always have to ask the question, is there a deeper insight here that I'm the one who's incorrect? And that's going to, as the systems get more and more intelligent, you're going to have to contend with that. It's like, is this a bug or a feature, what you just came up with?

Demis Hassabis

(00:59:14) Yeah. And they're going to be pretty complicated to do, but of course it will be, you can imagine also AI systems that are producing that code or whatever that is, and then human programmers looking at it, but also not unaided with the help of AI tools as well. So it's going to be kind of an interesting, maybe different AI tools to the ones the monitoring tools are the ones that generated it.

Lex Fridman

(00:59:36) So if we look at that AGI system, sorry to bring it back up, but AlphaEvolve, it's super cool. So AlphaEvolve enables on the programming side, something like recursive self-improvement potentially. If you can imagine what that AGI system, maybe not the first version, but a few versions beyond that, what does that actually look like? Do you think it'll be simple? Do you think it'll be something like a self-improving-

Lex Fridman

(01:00:00) Like, do you think it'll be simple? Do you think it'll be something like a self-improving program and a simple one?

Demis Hassabis

(01:00:06) I mean potentially that's possible. I would say I'm not sure it's even desirable because that's a kind of hard takeoff scenario. But these current systems like Alpha Evolve, they have human in the loop deciding on various things, there're separate hybrid systems that interact.

(01:00:22) One could imagine eventually doing that end to end. I don't see why that wouldn't be possible, but right now I think the systems are not good enough to do that in terms of coming up with the architecture of the code. And again, it's a little bit reconnected to this idea of coming up with a new conjectural hypothesis, how they're good if you give them very specific instructions about what you're trying to do, but if you give them a very vague high level instruction, that wouldn't work currently. And I think that's related to this idea of invent a game as good as Go, right?

(01:00:55) Imagine that was the prompt. That's pretty. And so the current systems wouldn't know I think what to do with that, how to narrow that down to something tractable. And I think there's similar, look, just make a better version of yourself. That's too unconstrained. But we've done it. And as you know with AlphaVol, like things like faster matrix multiplication, so when you hone it down to very specific thing you want, it's very good at incrementally improving that.

(01:01:21) But at the moment these are more incremental improvements, sort of small iterations. Whereas if you wanted a big leap in understanding, you'd need a much larger advance.

Lex Fridman

(01:01:34) Yeah. But it could also be sort of the pushback against hard takeoff scenario. It could be just a sequence of incremental improvements, like matrix multiplication. It has to sit there for days thinking how to incrementally improve a thing and it does solve recursively. And as you do more and more improvement, it'll slow down.

(01:01:55) So there be, the path to AGI won't be like a gradual improvement over time.

Demis Hassabis
(01:02:03) Yes. If it was just incremental improvements, that's how it would look. So the question is, could it come up with a new leap like the Transformers architecture? Could it have done that back in 2017 when we did it and Brain did it? And it's not clear that these systems, something our AlphaVol wouldn't be able to do, make such a big leap. So for sure these systems are good. We have systems I think that can do incremental hill climbing, and that's a kind of bigger question about is that all that's needed from here, or do we actually need one or two more big breakthroughs.

Lex Fridman
(01:02:34) And can the same kind of systems provide the breakthroughs also? So make it a bunch of S-curves like incremental improvement, but also every once in a while, leaps.

Demis Hassabis
(01:02:44) Yeah, I don't think anyone has systems that can have shown, unequivocally those big leaps that we have a lot of systems that do the hill climbing of the S-curve that you're currently on.

Lex Fridman
(01:02:55) And that would be the move 37 is a leap.

Demis Hassabis
(01:02:59) Yeah, I think it would be a leap, something like that.

Scaling laws

Lex Fridman
(01:03:01) Do you think the scaling laws are holding strong on the pre-training/post-training test time compute? Do you on the flip side of that, anticipate AI progress hit-

ting a wall?

Demis Hassabis

(01:03:13) We certainly feel there's a lot more room just in the scaling. So actually all steps pre-training, post-training, and inference time. So there's sort of three scalings that are happening concurrently. And again there, it's about how innovative you can be and we pride ourselves on having the broadest and deepest research bench. We have amazing, incredible researchers and people like Noam Shazir who came up with Transformers and Dave Silver who led the AlphaGo project and so on.

(01:03:47) And that research base means that if some new breakthrough is required, like an AlphaGo or Transformers, I would back us to be the place that does that. So I'm actually quite like it when the terrain gets harder, right? Because then it veers more from just engineering to true research, and research plus engineering, and that's our sweet spot and I think that's harder. It's harder to invent things than to fast follow.

(01:04:17) And so we don't know, I would say it's kind of 50/50 whether new things are needed or whether the scaling the existing stuff is going to be enough. And so, in true kind of empirical fashion, we are pushing both of those as hard as possible. The new blue sky, ideas and maybe about half our resources are on that. And then scaling to the max, the current capabilities. And we're still seeing some fantastic progress on each different version of Gemini.

Lex Fridman

(01:04:48) That's interesting the way you put it in terms of the deep bench, that if progress towards AGI is more than just scaling compute, so the engineering side of the problem, and is more on the scientific side where there's breakthroughs needed, then you feel confident DeepMind as well, Google DeepMind as well positioned to kick ass in that domain.

Demis Hassabis

(01:05:13) Well, I mean if you look at the history of the last decade or 15 years, it's been maybe, I don't know, 80-90% of the breakthroughs that underpins modern AI field today was from originally Google Brain, Google Research and DeepMind. So yeah, I would back that to continue hopefully.

Lex Fridman

(01:05:31) So on the data side, are you concerned about running out of high quality data, especially high quality human data?

Demis Hassabis
(01:05:37) I'm not very worried about that. Partly because I think there's enough data, and it's been proven to get the systems to be pretty good. And this goes back to simulations again. Do you have enough data to make simulations, so that you can create more synthetic data that are from the right distribution? Obviously that's the key. So you need enough real-world data in order to be able to create those kinds of data generators, and I think that we're at that step at the moment.

Lex Fridman
(01:06:05) Yeah, you've done a lot of incredible stuff on the side of science and biology, doing a lot with not so much data.

Demis Hassabis
(01:06:12) Yeah.

Lex Fridman
(01:06:12) I mean it's still a lot of data, but I guess enough to-

Demis Hassabis
(01:06:15) To get that going. Exactly. Exactly

Compute

Lex Fridman
(01:06:18) Yeah. How crucial is the scaling of compute to building AGI? That's an engineering question. It's almost a geopolitical question because it also integrated into that is supply chains and energy. A thing that you care a lot about, which is potentially fusion. So innovating on the side of energy also. Do you think we're going to keep scaling compute?

Demis Hassabis
(01:06:42) I think so, for several reasons. I think compute, there's the amount of compute you have for training, often it needs to be co-located, so actually even bandwidth constraints between data centers can affect that. So there's additional constraints even

there and that's important for training, obviously the largest models you can, but there's also because now AI systems are in products and being used by billions of people around the world, you need a ton of inference compute now.

(01:07:10) And then on top of that there's the thinking systems, the new paradigm of the last year that where they get smarter, the longer amount of inference time you give them at test time. So all of those things need a lot of compute and I don't really see that slowing down, and as AI systems become better, they'll become more useful and there'll be more demand for them. So both from the training side, the training side actually is only just one part of that. It may even become the smaller part of what's needed in the overall compute that's required.

Lex Fridman

(01:07:42) Yeah, that's one sort of almost meme-y kind of thing, which is the success in the incredible aspects of VL3. People kind of make fun of the more successful it becomes, the servers are sweating.

Demis Hassabis

(01:07:55) Yes.

Lex Fridman

(01:07:55) The inference.

Demis Hassabis

(01:07:57) Yeah, yeah, exactly. We did a little video of the servers frying eggs and things. That's right. And we are going to have to figure out how to do that. There's a lot of interesting hardware innovations that we do as we have our own TPU line and we're looking at inference-only things, inference-only chips and how we can make those more efficient.

(01:08:15) We're also very interested in building AI systems and we have done the help with energy usage, so help data center energy like for the cooling systems be efficient, grid optimization, and then eventually things like helping with plasma-containment fusion reactors. We've done lots of work on that with Commonwealth Fusion, and also one could imagine reactor design.

(01:08:38) And then material design I think is one of the most exciting. New types of solar material, solar panel material room temperature superconductors has always been

on my list of dream breakthroughs, and optimal batteries. And I think a solution to any one of those things would be absolutely revolutionary for climate and energy usage. And we're probably close, and again in the next five years to having AI systems that can materially help with those problems.

Future of energy

Lex Fridman
(01:09:05) If you were to bet, sorry for the ridiculous question, but what is the main source of energy in 20, 30, 40 years. Do you think it's going to be nuclear fusion?

Demis Hassabis
(01:09:15) I think fusion and solar are the two that I would bet on. Solar, I mean it's the fusion reactor in the sky of course, and I think really the problem there is batteries and transmission. So as well as more efficient, more and more efficient solar material perhaps eventually in space, these kind of Dyson Sphere type ideas.

(01:09:36) And fusion I think is definitely doable, it seems, if we have the right design of reactor and we can control the plasma and fast enough and so on, and I think both of those things will actually get solved. So we'll probably have at least those are probably the two primary sources of renewable, clean, almost free or perhaps free energy.

Lex Fridman
(01:09:58) What a time to be alive. If I traveled into the future with you a hundred years from now, how much would you be surprised if we've passed a type one Kardashev scale civilization?

Demis Hassabis
(01:10:11) I would not be that surprised if it was a hundred-year timescale from here. I mean I think it's pretty clear if we crack the energy problems in one of the ways we've just discussed or very efficient solar, then if energy is kind of free and renewable and clean, then that solves a whole bunch of other problems.

(01:10:32) So for example, the water access problem goes away because you can just use desalination. We have the technology, it's just too expensive. So only fairly wealthy countries like Singapore and Israel and so on actually use it. But if it was cheap, then all countries that have a coast could, but also you'd have unlimited rocket fuel. You could

just separate seawater out into hydrogen and oxygen using energy and that's rocket fuel.

(01:10:57) So combined with Elon's, amazing self landing rockets, then it could be you sort of like a bus service to space. So that opens up incredible new resources and domains. Asteroid mining I think will become a thing, and maximum human flourishing to the stars. That's what I dream about as well is like Carl Sagan's sort of idea of bringing consciousness to the universe, waking up the universe. And I think human civilization will do that in the full sense of time if we get AI right, and crack some of these problems with it.

Lex Fridman
(01:11:30) Yeah, I wonder what it would look like if you're just a tourist flying through space. You would probably notice earth because if you solve the energy problem, you would see a lot of space rockets probably. So it would be traffic here in London, but in space.

Demis Hassabis
(01:11:46) Yes, exactly.

Lex Fridman
(01:11:46) It's just a lot of rockets. And then you would probably see floating in space, some kind of source of energy like solar potentially. So earth would just look more on the surface, more technological. And then you would use the power of that energy then to preserve the natural...

Demis Hassabis
(01:12:05) Yes.

Lex Fridman
(01:12:06) Like the rainforest and all that kind of stuff.

Demis Hassabis
(01:12:07) Exactly. Because for the first time in human history we wouldn't be resource constrained. And I think that could be amazing new era for humanity where it's not zero-sum, right? I have this land, you don't have it. Or if the tigers have their forest, then the local villages can't, what are they going to use? I think that this will help a lot. No, it won't solve all problems because there's still other human foibles that will still ex-

ist, but it will at least remove one, I think one of the big vectors, which is scarcity of resources, including land and more materials and energy.

(01:12:45) And we should be sometimes call it another call about this kind of radical abundance era, where there's plenty of resources to go around. Of course the next big question is making sure that that's fairly, shared fairly and everyone in society benefits from that.

Human nature

Lex Fridman
(01:13:01) So there is something about human nature where I go, its like Borat, like my neighbor. You start trouble. We do start conflicts and that's why games throughout, as I'm learning actually more and more, even in ancient history, serve the purpose of pushing people away from war, actually hot war. So maybe we can figure out increasingly sophisticated video games that pull us, they give us that... Scratch the itch of conflict, whatever that is, but us, the human nature.

Demis Hassabis
(01:13:38) Like... Yeah.

Lex Fridman
(01:13:38) And then avoid the actual hot wars that would come with increasingly sophisticated technologies because we're now, we've long passed the stage where the weapons we're able to create can actually just destroy all of human civilization. So that's no longer a great way to start with your neighbor. It's better to play a game of chess.

Demis Hassabis
(01:14:03) Or football.

Lex Fridman
(01:14:03) Or football. Yeah.

Demis Hassabis
(01:14:05) And I think that's what my modern sport is and I love football watching it and I just feel like, and I used to play it a lot as well, and it's very visceral in its tribal, and I

think it does channel a lot of those energies into which I think is a kind of human need to belong to some group, but into a fun way, a healthy way and not destructive way, kind of constructive thing.

(01:14:33) And I think going back to games again is I think they're originally why they're so great as well for kids to play things like chess is they're great little microcosm simulations of the world. They're simulations of the world too. They're simplified versions of some real world situation, whether it's poker or Go or chess, different aspects or diplomacy, different of the real world.

(01:14:53) And it allows you to practice at them too, because how many times do you get to practice a massive decision moment in your life? What job to take, what university to go to? You get maybe, I don't know, a dozen or so key decisions one has to make and you've got to make those as best as you can. And games is a kind of safe environment, repeatable environment where you can get better at your decision-making process, and it maybe has this additional benefit of channeling some energies into more creative and constructive pursuits.

Lex Fridman
(01:15:24) Well I think it's also really important to practice losing and winning.

Demis Hassabis
(01:15:28) Right.

Lex Fridman
(01:15:29) Losing is a really, that's why I love games. That's why I love even things like Brazilian Jiu-Jitsu where you can get your kicked in a safe environment over and over. It reminds you about physics, about the way the world works about sometimes you lose, sometimes you win, you can still be friends with everybody. But that feeling of losing, I mean it's a weird one for us humans to really make sense of. That's just part of life. That is a fundamental part of life is losing.

Demis Hassabis
(01:16:00) And I think the martial arts as I understand it, but also in things like light chess is at least the way I took it's a lot to do with self-improvement, self-knowledge. That, okay, so I did this thing. It's not about really beating the other person, it's about maximizing your own potential.

(01:16:16) If you do it in a healthy way, you learn to use victory and losses in a way. Don't get carried away with victory and think you're just the best in the world. And the losses keep you humble, and always knowing there's always something more to learn. There's always a bigger expert that you can mentor you. I think you learn that I'm pretty sure in martial arts.

(01:16:35) And I think that's also the way that at least I was trained in chess. And so, in the same way, and it can be very hardcore and very important and of course you want to win, but you also need to learn how to deal with setbacks in a healthy way, and wire that feeling that you have when you lose something into a constructive thing of, next time I'm going to improve this or get better at this.

Lex Fridman
(01:16:57) There is something that's a source of happiness, a source of meaning that improvements that... It's not about the winning or losing.

Demis Hassabis
(01:17:04) Yes, the mastery. There's nothing more satisfying in a way. It's like, oh wow, this thing I couldn't do before. Now I can. And again, games and physical sports and mental sports, their ways of measuring their beautiful, because you can measure that progress.

Lex Fridman
(01:17:19) Yeah, there's something about I guess why I love role-playing games, like the number go up of on the skill tree, literally that is a source of meaning for us humans, whatever our-

Demis Hassabis
(01:17:29) Yeah, we're quite addicted to this sort of, these numbers going up. And maybe that's why we made games like that because obviously that is something we're hill climbing systems ourselves, right?

Lex Fridman
(01:17:42) Yeah. It would be quite sad if we didn't have any mechanism-

Demis Hassabis
(01:17:45) Color belts, we do this everywhere, where we just have thing that...

Lex Fridman

(01:17:51) And I don't want to dismiss that. There is a source of deep meaning across humans.

Google and the race to AGI

(01:17:55) So one of the incredible stories on the business, on the leadership side is what Google has done over the past year. So I think it's fair to say that Google was losing on the LLM product side a year ago with Gemini 1.5 And now it's winning, which... I'm Joe Biden. And you took the helm and you led this effort. What did it take to go from let's say quote-unquote losing to quote-unquote winning, in the span of a year?

Demis Hassabis
(01:18:22) Yeah, well firstly it's absolutely incredible team that we have led by Corey and Jeff Dean and Oriole and the amazing team we have on Gemini. Absolutely. So you can't do it without the best talent. And of course we have a lot of great compute as well. But then it's the research culture we've created and basically coming together both different groups in Google that was Google Brain, World-class team, and then the old DeepMind, and pulling together all the best people and the best ideas and gathering around to make the absolute greater system we could.

(01:18:59) And it was been hard, but we're all very competitive and we love research. This is so fun to do, and it's great to see our trajectory. It wasn't a given, but we're very pleased with where we are and the rate of progress is the most important thing. So if you look at where we've come to from two years ago to one year ago to now, I think we call it relentless progress. Along with relentless shipping of that progress is being very successful and it's unbelievably competitive, the whole space, the whole AI space, with some of the greatest entrepreneurs and leaders and companies in the world, all competing now because everyone's realized how important AI is. And it's very been pleasing for us to see that progress.

Lex Fridman
(01:19:47) Google's a gigantic company. Can you speak to the natural things that happen in that case is the bureaucracy that emerges? You want to be careful the natural, there's meetings and there's managers and that. What are some of the challenges from a leadership perspective, breaking through that in order to, like you said, ship? Like the number of products, Gemini related products that has been shipped over the past

years is insane.

Demis Hassabis

(01:20:14) Right? Yeah, exactly. That's what relentlessness looks like. I think it's a question of any big company ends up having a lot of layers of management and things like that is sort of the nature of how it works. But I still operate and I was always operating with old DeepMind as a start-up still. A large one, but still as a start-up.

(01:20:37) And that's what we still act like today with Google DeepMind. And acting with decisiveness and the energy that you get from the best smaller organizations. And we try to get the best of both worlds where we have this incredible, billions of users surfaces and credible products that we can power up with our AI and our research and that's amazing and that's very few places in the world you can get that, do incredible world-class research on the one hand and then plug it in and improve billions of people's lives the next day. That's a pretty amazing combination.

(01:21:10) And we're continually fighting and cutting away bureaucracy to allow the research culture and the relentless shipping culture to flourish. And I think we've got a pretty good balance, whilst being responsible with it, as you have to be as a large company and also with a number of huge product surfaces that we have.

Lex Fridman

(01:21:30) So a funny thing you mentioned about the surface with the billion, I had a conversation with a guy named, brilliant guy here at the British Museum, called Irvin Finkel. He's a world expert at cuneiforms, which is a ancient writing on tablets and he doesn't know about ChatGPT or Gemini, he doesn't even know about AI, but this first encounter with this AI is AI mode on Google.

Demis Hassabis

(01:21:57) Yes.

Lex Fridman

(01:21:58) He's like, is that what you're talking about, this AI mode? And it's just a reminder that there's a large part of the world that doesn't know about this AI thing.

Demis Hassabis

(01:22:08) Yeah, I know. It's funny. If you live on X and Twitter and I mean it's sort of at least my feed, it's all AI. And there's certain places where in the valley and certain

pockets where everyone's just, all they're thinking about is AI, but a lot of the normal world hasn't come across it yet.

Lex Fridman

(01:22:24) And that's a great responsibility to their first interaction. The grand scale of the rural, India or anywhere across the world you get to...

Demis Hassabis

(01:22:34) And we want it to be as good as possible and in a lot of cases it's just under the hood powering, making something like maps or search work better. And ideally for a lot of those people should just be seamless. It's just new technology that makes their lives more productive and helps them.

Lex Fridman

(01:22:50) A bunch of folks on the Gemini product and engineering teams spoken extremely highly of you on another dimension, that I almost didn't even expect. I kind of think of you as the deep scientists and caring about these big research scientific questions. But they also said you're a great product guy, like how to create a thing that a lot of people would use and enjoy using. So can you maybe speak to what it takes to create a AI based product that a lot of people enjoy using?

Demis Hassabis

(01:23:18) Yeah. Well, I mean, again, that comes back from my game design days where I used to design games for millions of gamers. People would forget about that. I've had experience with cutting edge technology in product that is how games was in the nineties.

(01:23:31) And so I love actually the combination of cutting edge research and then being applied in a product and to power a new experience. And so, I think it's the same skill really of imagining what it would be like to use it viscerally, and having good taste coming back to earlier. The same thing that's useful in science, I think can also be useful in product design.

(01:23:57) And I've just had a very, always been a sort of multidisciplinary person, so I don't see the boundaries really between arts and sciences, or product and research. It's a continuum for me. I like working on products that are cutting edge. I wouldn't be able to have cutting edge technology under the hood. I wouldn't be excited about them if

they were just run-of-the-mill products. It requires this invention, creativity, cap capability.

Lex Fridman

(01:24:23) What are some specific things you learned about when you, even on the LLM side, you're interacting with Gemini? This doesn't feel like, the layout, the interface, maybe the trade-off between the latency, how to present to the user, how long to wait and how that waiting is shown or the reason capabilities. There are some interesting things because like you said, it's the very cutting edge. We don't know how to present it correctly. So is there some specific things you've learned?

Demis Hassabis

(01:24:55) I mean it's such a false evolving space, evaluating this all the time, but where we are today is that you want to continually simplify things, whether that's the interface or what you build on top of the model, you kind of want to get out of the way of the model. The model train is coming down the track and it's improving unbelievably fast. This relentless progress we talked about earlier.

(01:25:17) You look at 2.5 versus 1.5 and it's just a gigantic improvement, and we expect that again for the future versions. And so the models are becoming more capable.

(01:25:26) So you've got, the interesting thing about the design space in today's world, these AI first products is you've got to design not for what the thing can do today, the technology can do today, but in a year's time. So you actually have to be a very technical product person, because you've got to have a good intuition for and feel for, okay, that thing that I'm dreaming about now can't be done today, but is the research track on schedule to basically intercept that in six months or a year's time.

(01:25:55) So you've kind of got to intercept where this highly changing technology's going, as well as the new capabilities are coming online all the time that we didn't realize before that can allow these research to work. Or now we've got video generation, what do we do with that, this multimodal stuff.

(01:26:13) Is it, one question I have is it really going to be the current UI that we have today, these text box chats? Seems very unlikely once you think about these super multimodal systems. Shouldn't it be something more like Minority Report where you are sort of vibing with it in a kind of collaborative way? It seems very restricted today. I

think we'll look back on today's interfaces and products and systems as quite archaic in maybe in just a couple of years.

(01:26:41) So I think there's a lot of space actually for innovation to happen on the product side as well as the research side.

Lex Fridman

(01:26:47) And then we are offline talking about the keyboard is, the open question is how, when and how much will we move to audio as the primary way of interacting with the machines around us versus typing stuff?

Demis Hassabis

(01:27:00) Yeah, I mean typing is a very low bandwidth way of doing it, even if you're a very fast typer. And I think we're going to have to start utilizing other devices, whether that's smart glasses, audio earbuds, and eventually maybe some sorts of neural devices, where we can increase the input and the output bandwidth to something maybe a 100x of what is today.

Lex Fridman

(01:27:24) I think that underappreciated art form is the interface design because I think you can not unlock the power of the intelligence of a system if you don't have the right interface. The interface is really the way you unlock its power. It's such an interesting question of how to do that. So how you would think getting out of the way isn't real art form.

Demis Hassabis

(01:27:46) Yes. It's the sort of thing that I guess Steve Jobs always talked about, right? It's simplicity, beauty, and elegance that we want. And we're not that nobody's there yet, in my opinion. And that's what I would like us to get to.

(01:27:58) Again, it sort of speaks to Go again as a game, the most elegant, beautiful game. Can you make an interface as beautiful as that? Actually, I think we're going to enter an era of AI-generated interfaces that are probably personalized to you, so it fits the way that you, your aesthetic, your feel, the way that your brain works and the AI kind of generates that depending on the task. That feels like that's probably the direction we'll end up in.

Lex Fridman

(01:28:25) Because some people are power users and they want every single parameter on the screen, everything based perhaps me with a keyboard-based navigation and to have shortcuts for everything. And some people like the minimalism.

Demis Hassabis
(01:28:37) Just hide all of that complexity. Yeah, exactly.

Lex Fridman
(01:28:39) Yeah. Well, I'm glad you have a Steve Jobs mode in you as well. This is great. Einstein mode, Steve Jobs mode.

(01:28:47) All right, let me try to trick you into answering a question. When will Gemini 3 come up? Is it before or after DTS-6? The world waits for both.

(01:28:56) And what does it take to go from 2.5 To 3.0? Because it seems like there's been a lot of releases of 2.5, which are already leaps in performance. So what does it even mean to go to a new version? Is it about performance? Is it about a completely different flavor of an experience?

Demis Hassabis
(01:29:16) Yeah, well, so the way it works with our different version numbers is we try to collect, so maybe it takes roughly six months or something to do a new kind of full run and the full productization of a new version.

(01:29:32) And during that time, lots of new interesting research iterations and ideas come up, and we sort of collect them all together that you could imagine the last six months worth of interesting ideas on the architecture front, maybe it's on the data front, it's like many different possible things. And we package that all up, test which ones are likely to be useful for the next iteration, and then bundle that all together. And then we start the new giant hero training run. And then of course that gets monitored.

Demis Hassabis
(01:30:00) ... run, right? And then of course that gets monitored and then at the end of the pre-training, then there's all the post-training, there's many different ways of doing that, different ways of patching it. So there's a whole experimental phase there which you can also get a lot of gains out. And that's where you see the version numbers usually referring to the base model, the pre-trained model, and then the interim versions of 2.5 and the different sizes and the different little additions. They're often patches or

post-training ideas that can be done afterwards off the same basic architecture. And then of course on top of that, we also have different sizes, Pro and Flash and Flashlight that are often distilled from the biggest ones, the Flash model from the Pro model. And that means we have a range of different choices. If you're the developer, do you want to prioritize performance or speed and cost?

(01:30:51) And we like to think of this Pareto frontier of on the one hand, the Y-axis is like performance, and then the X- axis is cost or latency and speed basically. And we have models that completely define the frontier. So whatever your trade-off is that you want as an individual user or as a developer, you should find one of our models satisfies that constraint.

Lex Fridman

(01:31:17) So behind the version changes, there is a big run and then there's just an insane complexity of productization. Then there's the distillation of the different sizes along that Pareto front. And then as with each step you take, you realize there might be a cool product. There's side quests.

Demis Hassabis

(01:31:39) Yes, exactly.

Lex Fridman

(01:31:41) And then you also don't want to take too many side quests because then you have a million versions and a million products.

Demis Hassabis

(01:31:45) Yes, precisely.

Lex Fridman

(01:31:46) It's very unclear, but you also get super excited because it's super cool. How does even look at VLs? Very cool. How does it fit into the bigger Thing?

Demis Hassabis

(01:31:55) Yes, exactly. Exactly. And then you're constantly this process of converging upstream, we call it ideas from the product surfaces or from the post-training and even further downstream and that, you upstream that into the core model training for the next run. So then the main model, the main Gemini track becomes more and more general and eventually, AGI.

Lex Fridman
(01:32:20) One hero run.

Demis Hassabis
(01:32:21) Yes, exactly. A few hero runs later.

Lex Fridman
(01:32:23) Yeah. So sometimes when you release these new versions or every version, really, are benchmarks productive or counterproductive for showing the performance of a model?

Demis Hassabis
(01:32:36) You need them, but it's important that you don't overfit to them. So they shouldn't be the be all and end all. So there's LMArena, or it used to be called LEMSYS, that's one of them that turned out organically to be one of the main ways people like to test these systems, at least the chatbots. Obviously there's loads of academic benchmarks that test mathematics and coding ability, general language ability, science ability and so on. And then we have our own internal benchmarks that we care about.

(01:33:04) It's a multi objective optimization problem. You don't want to be good at just one thing. We're trying to build general systems that are good across the board, and you try and make no-regret improvements. So where you improve in coding, but it doesn't reduce your performance in other areas. So that's the hard part because of course you could put more coding data in or you could put more, I don't know, gaming data in, but then does it make worse your language system or your translation systems and other things that you care about? So you've got to continually monitor this increasingly larger and larger suite of benchmarks. And also when you stick them into products, these models, you also care about the direct usage and the direct stats and the signals that you're getting from the end users, whether they're coders or the average person using the chat interfaces.

Lex Fridman
(01:34:00) Because ultimately, you want to measure the usefulness, but it's so hard to convert that into a number. It's really vibe based benchmarks across a large number of users. And it's hard to know and it would be just terrifying to me, you know have a much smarter model, but it's just something vibe based. It's not quite working. That's

such a scary and everything you just said. It has to be smart and useful across so many domains. So you get super excited all of a sudden solving programming problems you've never been able to solve before, but now it's crappier poetry or something and it's just, I don't know, that's a stressful. That's so difficult-

Demis Hassabis
(01:34:43) To balance.

Lex Fridman
(01:34:44) To balance and because you can't really trust the benchmarks, you really have to trust the end users.

Demis Hassabis
(01:34:48) Yeah. And then other things that are even more esoteric come into play, like the style of the persona of the system, is it verbose? Is it succinct? Is it humorous? And different people like different things. So it's very interesting. It's almost like cutting edge part of psychology research or personality research. I used to do that in my PhD, like five factor personality, what do we actually want our systems to be like? And different people will like different things as well. So these are all just new problems in product space that I don't think I've ever really been tackled before, but we're going to rapidly have to deal with now.

Lex Fridman
(01:35:27) I think it's a super fascinating space, developing the character of the thing and in so doing, it puts a mirror to ourselves, what are the kind of things that we like? Because prompt engineering allows you to control a lot of those elements, but can the product make it easier for you to control the different flavors of those experiences, the different characters that you interact with?

Demis Hassabis
(01:35:51) Yeah, exactly.

Competition and AI talent

Lex Fridman
(01:35:52) So what's the probability of Google DeepMind winning?

Demis Hassabis

(01:35:56) Well, I see it as winning. I think winning is the wrong way to look at it given how important and consequential what it is we're building. So funny enough, I try not to view it like a game or competition even though that's a lot of my mindset. It's about in my view, all of us or those of us at the leading edge or have a responsibility to steward this unbelievable technology that could be used for incredible good but also has risks, steward it safely into the world for the benefit of humanity. That's always what I've dreamed about and what we've always tried to do. And I hope that's what eventually the community, maybe the international community will rally around when it becomes obvious that as we get closer and closer to AGI, that's what's needed.

Lex Fridman

(01:36:43) I agree with you. I think that's beautifully put. You've said that you talk to and are on good terms with the leads of some of these labs. As the competition heats up, how hard is it to maintain those relationships?

Demis Hassabis

(01:36:58) It's been okay so far. I try to pride myself in being collaborative. I'm a collaborative person. Research is a collaborative endeavor. Science is a collaborative endeavor. It's all good for humanity in the end if you cure terrible diseases and you come up with an incredible cure, this is net win for humanity. And the same with energy, all of the things that I'm interested in helping solve with AI. So I just want that technology to exist in the world and be used for the right things and the benefits of that, the productivity benefits of that being shared for the benefit of everyone. So I try to maintain good relations with all the leading lab people. They're very interesting characters, many of them as you might expect.

Lex Fridman

(01:37:40) Yep.

Demis Hassabis

(01:37:41) But yeah, I'm on good terms I hope with pretty much all of them. And I think that's going to be important when things get even more serious than they are now, that there are those communication channels and that's what will facilitate cooperation or collaboration if that's what is required, especially on things like safety.

Lex Fridman

(01:38:00) Yeah, I hope there's some collaboration on stuff that's less high stakes and in so doing, serves as a mechanism for maintaining friendships and relationships. So for example, I think the internet would love it if you and Elon somehow collaborate on creating a video game, that kind of thing. I think that enables camaraderie and good terms. And also you two are legit gamers, so it's just fun to to create some-

Demis Hassabis

(01:38:22) Yeah, that would be awesome. And we've talked about that in the past and it may be a cool thing that we can do. And I agree with you, it'd be nice to have side projects in a way where one can just lean into the collaboration aspect of it and it's a a win-win for both sides and it builds up that collaborative muscle.

Lex Fridman

(01:38:44) I see the scientific endeavor as that side project for humanity and I think Google DeepMind has been really pushing that. I would love to see other labs do more scientific stuff and then collaborate because it just seems like it's easier to collaborate on the big scientific questions.

Demis Hassabis

(01:39:01) I agree and I would love to see a lot of people, all of the other labs talk about science, but I think we're really the only ones using it for science and doing that. And that's why projects like AlphaFold are so important to me. And I think to our mission is to show how AI can be clearly used in a very concrete way for the benefit of humanity. And also, we spun out companies like Isomorphic off the back of Alpha Fold to do drug discovery and it's going really well and you can think of build additional AlphaFold type systems to go into chemistry space to help accelerate drug design. And the examples I think we need to show and society needs to understand are where AI can bring these huge benefits.

Lex Fridman

(01:39:42) Well, from the bottom of my heart, thank you for pushing the scientific efforts forward with rigor, with fun, with humility, all of it. I just love to see and still talking about P equals NP, it's just incredible. So I love it. There's been seemingly a war for talent. Some of it is meme, I don't know. What do you think about Meta buying up talent with huge salaries and the heating up of this battle for talent? I should say that I think a

lot of people see DeepMind as a really great place to do cutting-edge work for the reasons that you've outlined. There's this vibrant scientific culture.

Demis Hassabis

(01:40:21) Yeah. Well look, of course there's a strategy that Meta is taking right now. I think that from my perspective at least, I think the people that are real believers in the mission of AGI and what it can do and understand the real consequences, both good and bad from that and what that responsibility entails, I think they're mostly doing it to be like myself, to be on the frontier of that research so they can help influence the way that goes and steward that technology safely into the world. And Meta right now are not at the frontier. Maybe they'll manage to get back on there and it's probably rational what they're doing from their perspective because they're behind and they need to do something. But I think there's more important things than just money. Of course one has to pay people their market rates and all of these things and that continues to go up. And I was expecting this because more and more people are finally realizing, leaders of companies, what I've always known for 30 plus years now, which is that AGI is the most important technology probably that's ever going to be invented. So in some senses, it's rational to be doing that. But I also think there's a much bigger question. People in AI these days are very well paid.

(01:41:32) I remember when we were starting out back in 2010, I didn't even pay myself a couple of years because it wasn't enough money. We couldn't raise any money, and these days, interns are being paid the amount that we raised as our first entire seed round. So it's pretty funny. And I remember the days where I used to have to work for free and almost pay my own way to do an internship. Right now, it's all the other around, but that's just how it is. It's the new world. But I think that we've been discussing what happens post- AGI and energy systems are solved and so on, what is even money going to mean? So I think in the economy and we're going to have much bigger issues to work through and how does the economy function in that world and companies? So I think it's a little bit of a side issue about salaries and things like that today.

Lex Fridman

(01:42:19) Yeah, when you're facing such gigantic consequences and gigantic, fascinating scientific questions-

Demis Hassabis

(01:42:25) Which may be only a few years away.

Future of programming

Lex Fridman

(01:42:27) So the practicals, the pragmatic sense, if we zoom in on jobs, we can look at programmers because it seems like AI systems are currently doing incredibly well at programming and increasingly so. So A lot of people that program for a living, love programming are worried they will lose their jobs. How worried should they be do you think, and what's the right way to adjust to the new reality and ensure that you survive and thrive as a human in the programming world?

Demis Hassabis

(01:42:58) Well, it's interesting that programming, and it's again counterintuitive to what we thought years ago, maybe that some of the skills that we think of as harder skills are turned out maybe to be the easier ones for various reasons. But coding and maths, because you can create a lot of synthetic data and verify if that data's correct. So because of that nature of that, it's easier to make things like synthetic data to train from. It's also an area of course we're all interested in because as programmers to help us and get faster at it and more productive.

(01:43:27) So I think for the next era, like the next five, 10 years, I think what we're going to find is people who embrace these technologies become almost at one with them, whether that's in the creative industries or the technical industries will become superhumanly productive, I think. So the great programmers will be even better, but there'll be even 10X even what they are today. And because there, you'll be able to use their skills to utilize the tools to the maximum, exploit them to the maximum. And so I think that's what we're going to see in the next domain. So that's going to cause quite a lot of change. And so that's coming. A lot of people benefit from that.

(01:44:05) So I think one example of that is if coding becomes easier, it becomes available to many more creatives to do more. But I think the top programmers will still have huge advantages as terms of specifying, going back to specifying what the architecture should be. The question should be how to guide these coding assistants in a way that's useful and check whether the code they produce is good. So I think there's plenty of headroom there for the foreseeable next few years.

Lex Fridman

(01:44:36) So I think there's several interesting things there. One is there's a lot of imperative to just get better and better consistently of using these tools so they're riding the wave of the improving models versus competing against them. But sadly, but that's the nature of life on earth, there could be a huge amount of value to certain kinds of programming at the cutting edge and less value to other kinds. For example, it could be front-end web design might be more amenable to, as you've mentioned, to generation by AI systems and maybe for example, game engine design or something like this or back-end design or guiding systems in high-performance situations, high-performance programming type of design decisions, that might be extremely valuable. But it will shift where the humans are needed most and that's scary for people to address.

Demis Hassabis

(01:45:37) Yeah, I think that's right. Anytime where there's a lot of disruption and change, and we've had this, it's not just this time. We've had this in many times in human history with the internet, mobile, but before that obviously, the Industrial Revolution and it's going to be one of those eras where there will be a lot of change. I think there'll be new jobs we can't even imagine today, just like the internet created. And then those people with the right skill sets to ride that wave will become incredibly valuable, those skills. But maybe people will have to relearn or adapt a bit, their current skills. And the thing that's going to be harder to deal with this time around is that I think what we're going to see is something like probably 10 times the impact the Industrial Revolution had, but 10 times faster as well. So instead of a 100 years, it takes 10 years and so that's going to make, it's like a 100X, the impact and the speed combined.

(01:46:31) So I think going to make it more difficult for society to deal with and there's a lot to think through and I think we need to be discussing that right now. And I encourage top economists in the world and philosophers to start thinking about how is society going to be affected by this and what should we do? Including things like universal basic provision or something like that where a lot of the increased productivity gets shared out and distributed to society and maybe in the form of services and other things where if you want more than that, you still go and get some incredibly rare skills and things like that and make yourself unique. But there's a basic provision that is provided.

Lex Fridman

(01:47:19) And if you think of government as a technology, there's also interesting questions, not just in the economics, but just politics. How do you design a system that's responding to the rapidly changing times such that you can represent the different pain that people feel from the different groups and how do you reallocate resources in a way that addresses that pain and represents the hope and the pain and the fears of different people in a way that doesn't lead to division? Because politicians are often really good at fueling the division and using that to get elected, defining the other and then saying that's bad. And based on that, I think that's often counterproductive to leveraging a rapidly changing technology to help the world flourish. So we almost need to improve our political systems as well rapidly, if you think of them as a technology.

Demis Hassabis

(01:48:19) Definitely. And I think we'll need new governance structures, institutions probably to help with this transition. So I think political philosophy and political science is going to be key to that. But I think the number one thing, first of all is to create more abundance of resources. So that's the number one thing. Increase productivity, get more resources, maybe eventually get out of the zero-sum situation. Then the second question is how to use those resources and distribute those resources. But yeah, you can't do that without having that abundance first.

John von Neumann

Lex Fridman

(01:48:54) You mentioned to me the book, The Maniac by Benjamin Labatut, a book on first of all about you. There's a bio about you.

Demis Hassabis

(01:49:05) Strange, yeah.

Lex Fridman

(01:49:06) Yes, sure. It's unclear how much is fiction, how much is reality. But I think the central figure that is John von Neumann, I would say it's a haunting and beautiful exploration of madness and genius and let's say the double-edged sword of discovery. And for people who don't know, John von Neumann is a legendary mind. He contributed to

quantum mechanics. He was on the Manhattan Project. He is widely considered to be the father of or pioneer the modern computer and AI and so on. So many people say he's one of the smartest humans ever, which is fascinating.

(01:49:45) And what's also fascinating is he's a person who saw nuclear science and physics become the atomic bomb, so you got to see ideas become a thing that has a huge amount of impact on the world. He also foresaw the same thing for computing, and that's a little bit again, beautiful and haunting aspect of the book. Then taking a leap forward and looking at this, at least it all AlphaZero, AlphaGo AlphaZero big moment that maybe John von Neumann's thinking was brought to reality. So I guess the question is what do you think if you got to hang out with John von Neumann now, what would he say about what's going on?

Demis Hassabis

(01:50:35) Well, that would be an amazing experience. He's a fantastic mind. And I also love the way he spent a lot of his time at Princeton at the Institute of Advanced Studies, a very special place for thinking. And it's amazing how much of a polymath he was and the spread of things he helped invent, including of course the Von Neumann architecture that all the modern computers are based on. And he had amazing foresight. I think he would've loved where we are today, and I think he would've really enjoyed AlphaGo being, he did game theory. I think he foresaw a lot of what would happen with learning machines, systems that are grown, I think he called it rather than programmed. I'm not sure how even maybe he wouldn't even be that surprised. There's the fruition of what I think he already foresaw in the 1950s.

Lex Fridman

(01:51:25) I wonder what advice he would give. He got to see the building of the atomic bomb with the Manhattan Project. I'm sure there's interesting stuff that maybe is not talked about enough, maybe some bureaucratic aspect, maybe the influence of politicians, maybe not enough of picking up the phone and talking to people that are called enemies by the said politicians. There might be some deep wisdom that we just may have lost from that time actually.

Demis Hassabis

(01:51:48) Yeah, I'm sure there is. I read a lot of books for that time as well, Chronicle Time and some brilliant people involved. But I agree with you. I think maybe there

needs to be more dialogue and understanding. I hope we can learn from those times. I think the difference here is that the AI has so many, it's a multi-use technology. Obviously we're trying to do things like solve all diseases, help with energy and scarcity, these incredible things. This is why all of us and myself, I started on this journey 30 plus years ago. But of course there are risks too. And probably Von Neumann, my guess is he foresaw both. And I think he said, I think it's to his wife, that computers would be even more impactful in the world. And as we just discussed, I think that's right. I think it's going to be 10 times at least of the Industrial Revolution. So I think he's right. So I think he would've been, I imagine, fascinated by where we are now.

Lex Fridman

(01:52:53) And I think one of the, maybe you can correct me, but one of the takeaways from the book is that reason, as said in the book, Mad Dreams of Reason, it's not enough for guiding humanity as we build these super powerful technology. That there's something else. There's also a religious component, whatever God, whatever religion gives, it pulls at something in the human spirit that raw cold reason doesn't give us.

Demis Hassabis

(01:53:22) And I agree with that. I think we need to approach it with whatever you want to call it, a spiritual dimension or humanist dimension. Doesn't have to be to do with religion, but this idea of a soul, what makes us human, this spark that we have, perhaps it's to do with consciousness when we finally understand that, I think that has to be at the heart of the endeavor. And technology, I've always seen technology as the enabler, the tools that enable us to flourish and to understand more about the world. And I'm with Feynman on this, and he used to always talk about science and art being companions. You can understand it from both sides, the beauty of a flower, how beautiful it is, and also understand why the colors of the flower evolve like that. That just makes it more beautiful, just the intrinsic beauty of the flower.

(01:54:10) I've always seen it like that. And maybe in the Renaissance times, the great discoverers then, people like Da Vinci, I don't think he saw any difference between science and art and perhaps religion. Everything was, it's just part of being human and being inspired about the world around us. And that's the philosophy I tried to take. And one of my favorite philosophers is Spinoza. And I think he combined that all very well, this idea of trying to understand the universe and understanding our place in it. And

that was his way of understanding religion. And I think that's quite beautiful. And for me, all of these things are related, interrelated, the technology and what it means to be human.

(01:54:53) And I think it's very important though that we remember that as when we're immersed in the technology and the research, I think a lot of researchers that I see in our field are a little bit too narrow and only understand the technology. And I think also that's why it's important for this to be debated by society at large. I'm very supportive of things like the AI summits that will happen and governments understanding it. And I think that's one good thing about the chatbot era and the product era of AI is that everyday person can actually feel and interact with cutting edge AI and feel it for themselves.

Lex Fridman
(01:55:30) Yeah, because they force the technologists to have the human conversation. Yeah, for sure.

Demis Hassabis
(01:55:30) Yeah.

Lex Fridman
(01:55:35) That's the hopeful aspect of it, like you said, it's a dual use technology that we're forcefully integrating the entire humanity into it, into the discussion about AI because ultimately AI, AGI will be used for things that states use technologies for, which is conflict and so on. And the more we integrate humans into this picture by having chats with them, the more we will guide.

Demis Hassabis
(01:56:01) Yeah, be able to adapt, society will be able to adapt to these technologies we've always done in the past with the incredible technologies we've invented in the past.

Lex Fridman
(01:56:10) Do you think there will be something like a Manhattan Project where there will be an escalation of the power of this technology and states in their old way of thinking, we'll try to use it as weapons technologies and there will be this escalation?

Demis Hassabis

(01:56:27) I hope not. I think that would be very dangerous to do. And I think also not the right use of the technology. I hope we'll end up with something more collaborative if needed, more like a CERN project where it's research-focused and the best minds in the world come together to carefully complete the final steps and make sure it's responsibly done before deploying it to the world. We'll see. It's difficult with the current geopolitical climate, I think, to see cooperation, but things can change. And I think at least on the scientific level, it's important for the researchers to keep in touch and keep close to each other at least on those kinds of topics.

Lex Fridman

(01:57:17) And I personally believe on the education side and immigration side, it would be great if both directions, people from the West immigrated to China and China, back. There is some family human aspect of people just intermixing and thereby those ties grow strong. So you can't divide against each other, this old school way of thinking. And so multicultural, multidisciplinary research teams working on scientific questions, that's like the hope. Don't let the leaders that are warmongers divide us. I think science is the ultimately really beautiful connector.

Demis Hassabis

(01:57:55) Yeah, science has always been, I think, quite a very collaborative endeavor and scientists know that it's a collective endeavor as well, and we can all learn from each other. So perhaps it could be a vector to get a bit of cooperation.

p(doom)

Lex Fridman

(01:58:08) Ridiculous question, what's your P-Doom? Probability of the human civilization destroys itself?

Demis Hassabis

(01:58:14) Well, look, I don't have a P-Doom number. The reason I don't is because I think it would imply a level of precision that is not there. So I don't know how people are getting their P-Doom numbers. I think it's a little bit of ridiculous notion because what I would say is it's definitely non-zero and it's probably non-negligible. So that in itself is pretty sobering. And my view is it's just hugely uncertain what these technologies are going to be able to do, how fast are they going to take off, how controllable are

they going to be. Some things may turn out to be, and hopefully way easier than we thought, but it may be there's some really hard problems that are harder than we guessed today, and I think we don't know that for sure. And so under those conditions of a lot of uncertainty, but huge stakes both ways.

(01:59:09) On the one hand, we could solve all diseases, energy problems, the scarcity problem, and then travel to the stars and conscious of the stars and maximum human flourishing. On the other hand, is these P-Doom scenarios. So given the uncertainty around it and the importance of it, it's clear to me the only rational, sensible approach is to proceed with cautious optimism. So we want the benefits of course, and all of the amazing things that AI can bring. And actually, I would be really worried for humanity given the other challenges that we have, climate, aging, resources, all of that if I didn't know something like AI was coming down the line. How would we solve all those other problems? I think it's hard. So I think it could be amazingly transformative for good. But on the other hand, there are these risks that we know are there.

Demis Hassabis

(02:00:00) But on the other hand, there are these risks that we know are there, but we can't quite quantify. So the best thing to do is to use the scientific method to do more research to try and more precisely define those risks and of course address them. And I think that's what we're doing. I think there probably needs to be 10 times more effort of that than there is now as we are getting closer and closer to the AGI line.

Lex Fridman

(02:00:27) What would be the source of worry for you more? Would it be human-caused or AI, AGI caused? Are humans abusing that technology versus AGI itself through mechanism that you've spoken about, which is fascinating, deception or this kind of stuff getting better and better and better secretly and then escapes?

Demis Hassabis

(02:00:45) I think they operate over different timescales and they're equally important to address. So there's just the common garden variety of bad actors using new technology, in this case, general purpose technology and repurposing it for harmful end. And that's a huge risk and I think that has a lot of complications because generally I'm in huge favor of open science and open source, and in fact, we did it with all our science projects like AlphaFold and all of those things for the benefit of the scientific communi-

ty. But how does one restrict bad actors access to these powerful systems, whether they're individuals or even rogue states, but enable access at the same time to good actors to maximally build on top of? It's pretty tricky problem that I've not heard a clear solution to. So there's the bad actor use case problem, and then there's obviously, as the systems become more agentic and closer to AGI and more autonomous, how do we ensure the guardrails and they stick to what we want them to do and under our control?

Lex Fridman
(02:01:52) Yeah, I tend to, maybe my mind is limited, worry more about the humans, so the bad actors. And there it could be in part how do you not put destructive technology in the hands of bad actors, but in another part from, again, geopolitical technology perspective, how do you reduce the number of bad actors in the world? That's also an interesting human problem.

Demis Hassabis
(02:02:14) Yeah, it's a hard problem. I mean, look, we can maybe also use the technology itself to help early warning on some of the bad actor use cases, right? Whether that's bio or nuclear or whatever it is, AI could be potentially helpful there as long as the AI that you're using is itself reliable, right? So it's a sort of interlocking problem and that's what makes it very tricky. And again, it may require some agreement internationally, at least between China and the U.S. of some basic standards. Right.

Humanity

Lex Fridman
(02:02:50) I have to ask you about the book, The Maniac. There's the hand of God moment, Lee Sedol's move 78 that perhaps the last time a human did a move of pure human genius and beat AlphaGo or broke its brain.

Demis Hassabis
(02:03:08) Yes.

Lex Fridman
(02:03:08) Sorry to anthropomorphize, but it's an interesting moment because I think in so many domains it will keep happening.

Demis Hassabis

(02:03:14) Yeah, it's a special moment and it was great for Lee Sedol. I think it's in a way they were inspiring each other. We as a team were inspired by Lee Sedol's brilliance and nobleness. Then maybe he got inspired by what AlphaGo was doing to then conjure this incredible inspirational moment, captured very well in the documentary about it. And I think that'll continue in many domains where there's this, at least again for the foreseeable future, of the humans bringing in their ingenuity and asking the right question, let's say, and then utilizing these tools in a way that then cracks a problem.

Lex Fridman

(02:03:58) Yeah. As the AI become smarter and smarter, one of the interesting questions we can ask ourselves is what makes humans special? It does feel perhaps biased that we humans are deeply special. I don't know if it's our intelligence, it could be something else, that other thing that's outside the mad dreams of reason.

Demis Hassabis

(02:04:20) I think that's what I've always imagined when I was a kid and starting on this journey of I was of course fascinated by things like consciousness, did a neuroscience PhD to look at how the brain works, especially imagination and memory. I focused on the hippocampus and it's sort of going to be interesting. I always thought the best way, of course, one can philosophize about it and have thought experiments and maybe even do actual experiments like you do in neuroscience on real brains. But in the end, I always imagine that building AI, a kind of intelligent artifact, and then comparing that to the human mind and seeing what the differences were would be the best way to uncover what's special about the human mind, if indeed there is anything special.

(02:05:00) And I suspect there probably is, but it's going to be hard to... I think this journey we're on will help us understand that and define that. And there may be a difference between carbon based substrates that we are and silicon ones when they process information. One of the best definitions I like of consciousness is it's the way information feels when we process it, right?

Lex Fridman

(02:05:22) Yeah.

Demis Hassabis

(02:05:24) It could be. I mean, it's not a very helpful scientific explanation, but I think it's kind of interesting intuitive one. And so on this journey, this scientific journey we're on will I think help uncover that mystery.

Lex Fridman
(02:05:36) Yeah. What I cannot create, I do not understand. That's somebody you deeply admire, Richard Feynman, like you mentioned. You also reach for the Wigner's dreams of universality that he saw in constrained domains, but also broadly generally in mathematics and so on. So many aspects on which you're pushing towards not to start trouble at the end, but Roger Penrose.

Consciousness and quantum computation

Demis Hassabis
(02:06:00) Yes. Okay.

Lex Fridman
(02:06:04) So do you think consciousness, there's this hard problem of consciousness, how information feels. Do you think consciousness, first of all, is a computation? And if is, if it's information processing, like you said, everything is, is it something that could be modeled by a classical computer?

Demis Hassabis
(02:06:23) Yeah.

Lex Fridman
(02:06:24) Or is it a quantum mechanical in nature?

Demis Hassabis
(02:06:26) Well, look, Penrose is an amazing thinker, one of the greatest of the modern era, and we've had a lot of discussions about this. Of course, we cordially disagree, which is I feel like... I mean, he collaborated with a lot of good neuroscientists to see if he could find mechanisms for quantum mechanics behavior in the brain. And to my knowledge, they haven't found anything convincing yet. So my betting is that it's mostly it is just classical computing that's going on in the brain, which suggests that all the phenomena are modelable or mimicable by a classical computer. But we'll see. There may be this final mysterious things of the feeling of consciousness, the qualia, these

kinds of things that philosophers debate where it's unique to the substrate.

(02:07:12) We may even come towards understanding that when if we do things like neural link or have neural interfaces to the AI systems, which I think we probably will eventually, maybe to keep up with the AI systems, we might actually be able to feel for ourselves what it's like to compute on silicon, right? And maybe that will tell us. So I think it's going to be interesting. I had a debate once with the late Daniel Dennett about why do we think each other are conscious? Okay, so it's for two reasons. One is you're exhibiting the same behavior that I am. So that's one thing. Behaviorally you seem like a conscious being if I am.

(02:07:49) But the second thing which is often overlooked is that we're running on the same substrate. So if you're behaving in the same way and we're running on the same substrate, it's most parsimonious to assume you're feeling the same experience that I'm feeling. But with an AI that's on silicon, we won't be able to rely on the second part, even if it exhibits the first part, that behavior looks like a behavior of a conscious being. It might even claim it is, but we wouldn't know how it actually felt and it probably couldn't know what we felt, at least in the first stages. Maybe when we get to superintelligence and the technologies that builds, perhaps we'll be able to bridge that.

Lex Fridman
(02:08:26) No, I mean that's a huge test for radical empathy is to empathize with a different substrate.

Demis Hassabis
(02:08:32) Right. Exactly. We've never had to confront that before.

Lex Fridman
(02:08:36) Yeah. So maybe through brain computer interfaces be able to truly empathize what it feels like to be a computer, to compute.

Demis Hassabis
(02:08:42) Well, for information to be computed not on a carbon system.

Lex Fridman
(02:08:46) I mean, that's deeply... Some people kind of think about that with plants, with other life forms which are different.

Demis Hassabis

(02:08:51) Yes, it could be exactly.

Lex Fridman

(02:08:53) Similar substrate, but sufficiently far enough on the evolutionary tree that it requires a radical empathy, but to do that with a computer.

Demis Hassabis

(02:09:02) I mean, look, there are animal studies on this. Of course, higher animals like killer whales and dolphins and dogs and monkeys, they have some, and elephants, they have some aspects certainly of consciousness, right? Even though they might not be that smart on an IQ sense. So we can already empathize with that and maybe even some of our systems one day, like we built this thing called DolphinGemma, which a version of our system was trained on dolphin and whale sounds, and maybe we'll be able to build an interpreter or translator at some point which would be pretty cool.

Lex Fridman

(02:09:35) What gives you hope for the future of human civilization?

Demis Hassabis

(02:09:38) Well, what gives me hope is that I think our almost limitless ingenuity, first of all. I think the best of us and the best human minds are incredible. And I love meeting and watching any human that's the top of their game, whether that's sport or science or art, it's just nothing more wonderful than that, seeing them in their element in flow. I think it's almost limitless. Our brains are general systems, intelligent systems, so I think it's almost limitless what we can potentially do with them. And then the other thing is our extreme adaptability. I think it's going to be okay in terms of there's going to be a lot of change, but look where we are now without effectively our hunter-gatherer brains.

(02:10:24) How is it we can cope with the modern world, right? Flying on planes, doing podcasts, playing computer games and virtual simulations. I mean, it's already mind blowing given that our mind was developed for hunting buffaloes on the tundra. And so I think this is just the next step, and it's actually kind of interesting to see how society's already adapted to this mind blowing AI technology we have today already. It's sort of like, "Oh, I talked to chat bots. Totally fine."

Lex Fridman

(02:10:54) And it's very possible that this very podcast activity, which I'm here for, will be completely replaced by AI. I'm very replaceable and I'm waiting for it.

Demis Hassabis
(02:11:02) Not to the level that you can do it, Lex, I don't think.

Lex Fridman
(02:11:04) Thank you. That's what we humans do to each other. We compliment.

Demis Hassabis
(02:11:08) Exactly.

Lex Fridman
(02:11:09) All right. And I'm deeply grateful for us humans to have this infinite capacity for curiosity, adaptability, like you said, and also compassion and ability to love.

Demis Hassabis
(02:11:18) Exactly.

Lex Fridman
(02:11:19) All of those human things.

Demis Hassabis
(02:11:19) All the things that are deeply human.

Lex Fridman
(02:11:21) Well, this is a huge honor, Demis. You are one of the truly special humans in the world. Thank you so much for doing what you do and for talking today.

Demis Hassabis
(02:11:29) Well, thank you very much, Lex.

Lex Fridman
(02:11:32) Thanks for listening to this conversation with Demis Hassabis. To support this podcast, please check out our sponsors in the description and consider subscribing to this channel. And now let me answer some questions and try to articulate some things I've been thinking about. If you'd like to submit questions including in audio and video form, go to lexfridman.com/ama. I got a lot of amazing questions, thoughts and requests from folks. I'll keep trying to pick some randomly and comment on it at the end of every episode. I got a note on May 21st this year that said, "Hi, Lex. 20 years ago

today, David Foster Wallace delivered his famous This is Water speech at Kenyon College. What do you think of this speech?

David Foster Wallace

(02:12:21) Well, first, I think this is probably one of the greatest and most unique commencement speeches ever given, but of course, I have many favorites, including the one by Steve Jobs. And David Foster Wallace is one of my favorite writers and one of my favorite humans. There's a tragic honesty to his work, and it always felt as if he was engaging in a constant battle with his own mind, and the writing, his writing were kind of his notes from the front lines of that battle. Now onto the speech, let me quote some parts. There's of course the parable of the fish and the water that goes, there are these two young fish swimming along and they happen to meet an older fish swimming the other way who nods at them and says, "Morning boys, how's the water?" And the two young fish swim on for a bit and then eventually one of them looks over at the other and goes, "What the hell is water?" In the speech, David Foster Wallace goes on to say, "The point of the fish story is merely that the most obvious important realities are often the ones that are hardest to see and talk about. Stated as an English sentence of course, this is just the banal platitude, but the fact is that in the day to day trenches of adult existence, banal platitudes can have a life or death importance, or so I wish to suggest to you in this dry and lovely morning." I have several takeaways from this parable and the speech that follows. First, I think we must question everything, and in particular, the most basic assumptions about our reality, our life, and the very nature of existence, and that this project is a deeply personal one. In some fundamental sense, nobody can really help you in this process of discovery.

(02:14:23) The call to action here, I think, from David Foster Wallace as he puts it, is to " To be just a little less arrogant, to have just a little more critical awareness about myself and my certainties because a huge percentage of the stuff that I tend to be automatically certain of is it turns out totally wrong and deluded." All right, back to me. Lex speaking. Second takeaway is that the central spiritual battles of our life are not fought on a mountain top somewhere at a meditation retreat, but it's fought in the mundane moments of daily life.

(02:15:08) Third takeaway is that we too easily give away our time and attention to the

multitude of distractions that the world feeds us, the insatiable black holes of attention. David Foster Wallace's call to action in this case is to be deeply aware of the beauty in each moment and to find meaning in the mundane. I often quote David Foster Wallace in his advice that the key to life is to be unborable, and I think this is exactly right. Every moment, every object, every experience when looked at closely enough contains within it infinite richness to explore. And since Demis Hassabis of this very podcast episode and I are such fans of Richard Feynman, allow me to also quote Mr. Feynman on this topic as well.

(02:16:04) "I have a friend who's an artist and has sometimes taken a view which I don't agree with very well. He'll hold up a flower and say, "Look how beautiful it is," and I'll agree. Then he says, "I as an artist can see how beautiful this is, but you as a scientist take this all apart and it becomes a dull thing," and I think that's kind of nutty. First of all, the beauty that he sees is available to other people and to me too, I believe. Although I may not be quite as refined aesthetically as he is, I can appreciate the beauty of a flower. At the same time, I see much more about the flower than he sees. I could imagine the cells in there, the complicated actions inside which also have beauty. I mean, it's not just beauty at this dimension at one centimeter, there's also beauty at the smaller dimensions."

(02:17:05) "Their inner structure, also the processes, the fact that the colors and the flower evolved in order to attract the insects to pollinate it is interesting. It means that the insects can see the color. It adds a question. Does this aesthetic sense also exist in lower forms? Why is it aesthetic? All kinds of interesting questions, which the science knowledge only adds to the excitement, the mystery, and the awe of a flower. It only adds."

(02:17:36) All right, back to David Foster Wallace's speech. He has a great story in there that I particularly enjoy. It goes, there are these two guys sitting together in a bar in the remote Alaskan wilderness. One of the guys is religious, the other is an atheist, and the two are arguing about the existence of God with that special intensity that comes after about the fourth beer. And the atheist says, "Look, it's not like I don't have actual reasons for not believing in God. It's not like I haven't ever experimented with the whole God and prayer thing. Just last month, I got caught away from the camp in that terrible blizzard, and I was totally lost and I couldn't see a thing and it was 50 below. So I tried

it. I fell in my knees in the snow and cried out, 'Oh God, if there is a God, I'm lost in this blizzard and I'm going to die if you don't help me."

(02:18:35) And now back in the bar, the religious guy looks at the atheist all puzzled, "Well, then you must believe now?" he says, "After all, there you are, alive." The atheist just rolls his eyes. "No, man. All that happened was a couple of Eskimos happened to be wandering by and showed me the way back to the camp." All this, I think, teaches us that everything is a matter of perspective and that wisdom may arrive if we have the humility to keep shifting and expanding our perspective on the world. Thank you for allowing me to talk a bit about David Foster Wallace. He's one of my favorite writers and he's a beautiful soul.

Education and research

(02:19:20) If I may, one more thing I wanted to briefly comment on. I find myself to be in this strange position of getting attacked online often from all sides, including being lied about sometimes through selective misrepresentation, but often through down-right lies. I don't know how else to put it. This all breaks my heart, frankly, but I've come to understand that it's the way of the internet and the cost of the path I've chosen. There's been days when it's been rough on me mentally. It's not fun being lied about, especially when it's about things that are usually for a long time have been a source of happiness and joy for me. But again, that's life.

(02:20:04) I'll continue exploring the world of people and ideas with empathy and rigor, wearing my heart on my sleeve as much as I can. For me, that's the only way to live. Anyway, a common attack on me is about my time at MIT and Drexel, two great universities I love and have tremendous respect for. Since a bunch of lies have accumulated online about me on these topics, to a sad and at times hilarious degree, I thought I would once more state the obvious facts about my bio for the small number of you who may care. TLDR, two things. First, as I say often, including in a recent podcast episode that somehow was listened to by many millions of people, I proudly went to Drexel University for my bachelor's, master's, and doctorate degrees.

(02:20:59) Second, I am a research scientist at MIT and have been there in a paid research position for the last 10 years. Allow me to elaborate a bit more on these two things now, but please skip if this is not at all interesting. So like I said, a common at-

tack on me is that I have no real affiliation with MIT. The accusation, I guess, is that I'm falsely claiming an MIT affiliation because I taught a lecture there once. Nope, that accusation against me is a complete lie. I have been at MIT for over 10 years in a paid research position from 2015 to today. To be extra clear, I'm a research scientist at MIT working in LIDS, the Laboratory for Information and Decision Systems in the College of Computing. For now, since I'm still at MIT, you can see me in the directory and on the various lab pages.

(02:22:05) I have indeed given many lectures at MIT over the years, a small fraction of which I posted online. Teaching for me always has been just for fun and not part of my research work. I personally think I suck at it, but I have always learned and grown from the experience. It's like Feynman spoke about, if you want to understand something deeply, it's good to try to teach it. But like I said, my main focus has always been on research. I published many peer-reviewed papers that you can see in my Google Scholar profile. For my first four years at MIT, I worked extremely intensively. Most weeks were 80 to 100-hour work weeks. After that, in 2019, I still kept my research scientist position, but I split my time taking a leap to pursue projects in AI and robotics outside MIT and to dedicate a lot of focus to the podcast.

(02:23:03) As I've said, I've been continuously surprised just how many hours preparing for an episode takes. There are many episodes of the podcast for which I have to read, write, and think for 100, 200 or more hours across multiple weeks and months. Since 2020, I have not actively published research papers. Just like the podcast, I think it's something that's a serious full-time effort. But not publishing and doing full-time research has been eating at me because I love research and I love programming and building systems that test out interesting technical ideas, especially in the context of human-AI or human-robot interaction. I hope to change this in the coming months and years.

(02:23:52) What I've come to realize about myself is if I don't publish or if I don't launch systems that people use, I definitely feel like a piece of me is missing. It legitimately is a source of happiness for me. Anyway, I'm proud of my time at MIT. I was and am constantly surrounded by people much smarter than me, many of whom have become lifelong colleagues and friends. MIT is a place I go to escape the world, to focus on exploring fascinating questions at the cutting edge of science and engineering. This, again,

makes me truly happy and it does hit pretty hard on a psychological level when I'm getting attacked over this. Perhaps I'm doing something wrong. If I am, I will try to do better.

(02:24:43) In all this discussion of academic work, I hope you know that I don't ever mean to say that I'm an expert at anything. In the podcast and in my private life, I don't claim to be smart. In fact, I often call myself an idiot and mean it. I try to make fun of myself as much as possible, and in general to celebrate others instead. Now to talk about Drexel University, which I also love, am proud of and am deeply grateful for my time there. As I said, I went to Drexel for my bachelor's, master's, and doctorate degrees in computer science and electrical engineering. I've talked about Drexel many times, including, as I mentioned, at the end of a recent podcast, the Donald Trump episode. funny enough, that was listened to by many millions of people where I answered a question about graduate school and explained my own journey at Drexel and how grateful I am for it.

(02:25:46) If it's at all interesting to you, please go listen to the end of that episode or watch the related clip. At Drexel, I met and worked with many brilliant researchers and mentors from whom I've learned a lot about engineering, science and life. There are many valuable things I gained from my time at Drexel. First, I took a large number of very difficult math and theoretical computer science courses. They taught me how to think deeply and rigorously, and also how to work hard and not give up even if it feels like I'm too dumb to find a solution to a technical problem.

(02:26:21) Second, I programmed a lot during that time, mostly C, C++. I programmed robots, optimization algorithms, computer vision systems, wireless network protocols, multimodal machine learning systems, and all kinds of simulations of physical systems. This is where I really developed a love for programming, including, yes, Emacs And the Kinesis keyboard. I also, during that time, read a lot, I played a lot of guitar, wrote a lot of crappy poetry, and trained a lot in judo and jiu-jitsu, which I cannot sing enough praises to. Jiu-jitsu humbled me on a daily basis throughout my twenties, and it still does to this very day whenever I get a chance to train.

(02:27:13) Anyway, I hope that the folks who occasionally get swept up in the chanting online crowds that want to tear down others don't lose themselves in it too much. In the end, I still think there's more good than bad in people. But we're all each of us a

mixed bag. I know I am very much flawed. I speak awkwardly. I sometimes say stupid shit. I can get irrationally emotional. I can be too much of a dick when I should be kind. I can lose myself in a biased rabbit hole before I wake up to the bigger, more accurate picture of reality. I'm human and so are you for better or for worse, and I do still believe we're in this whole beautiful mess together. I love you all.